

UNIT - I

FUNDAMENTAL OF MECHANICS

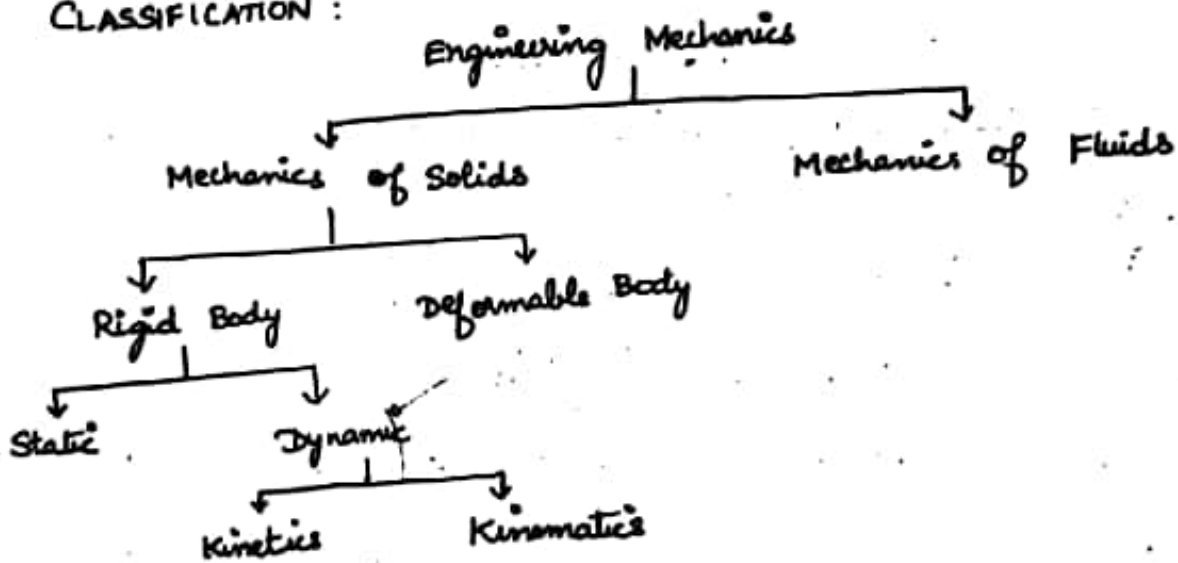
ENGINEERING MECHANICS:

It is defined as the branch of science which describes and predicts the condition of rest or motion of bodies under action of force.

APPLICATIONS OF ENGINEERING MECHANICS:

- Working principles of engines.
- External forces acting on a ship or boat in water.
- Study of missiles and aeroplane.
- Forces developed in trusses, frames to rest windows.

CLASSIFICATION:



RIGID BODY:

It is defined as the body which does not undergo any deformation, change in its shape and volume when external forces are applied on it.

PARTICLE:

A body of negligible dimension is called a particle.

STATICS:

It is the branch of science which deals with the study of a body at rest.

eg: (1) Equilibrium of book lying on the table and study of different forces acting on it.

(2) Support Reactions of beam subjected to some external load

DYNAMICS:

It is the branch of which deals with the study of a body in motion.

eg: (1) Force hit by a cricket bat on ball

(2) Force applied on brakes when the moving car is brought to rest

KINEMATICS:

It is the study of a body in motion without considering the forces that cause the motion.

eg: Displacement, velocity, acceleration.

KINETICS:

It is the study of a body in motion with considering the forces that cause the motion.

eg: Forces acting on the body, Mass of the body.

FORCE:

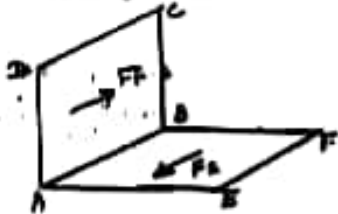
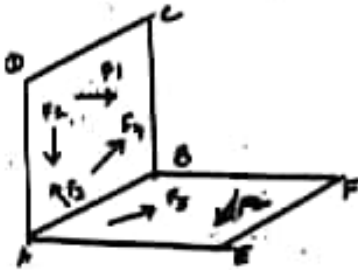
It is defined as the action of one body on another which affects the state of rest or motion of the body acted on it. It is a vector quantity. Its unit is Newton (N) or (kg m/s²)

CHARACTERISTICS OF FORCE:

- a) Direction
- b) Magnitude
- c) Nature of force
- d) Line of action.

TYPES OF FORCE SYSTEM :

S.No	Force System	Figure	Characteristics
1.	COPLANAR FORCE		Line of action of all forces lie in the same plane.
2.	COPLANAR - COLINEAR		Line of action of all forces lying on same plane and also acting along same line.
3.	COPLANAR - CONCURRENT		Line of action of all forces lying on same plane and also passing through a same point.
4.	COPLANAR - PARALLEL		Line of action of all forces lying on same plane and also parallel to each other.
5.	COPLANAR NON-CONCURRENT NON-PARALLEL		Line of action of all forces lying on same plane and does not pass through same point and also not parallel to each other.
6.	NON-COPLANAR FORCE		Line of action of all forces do not lie on the same plane but lie on different planes.
7.	NON-COPLANAR CONCURRENT		Line of action of all forces not lying on the same plane but passes through same point.

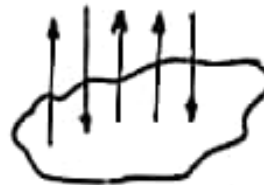
SNO	Force System	Figure	Characteristics
8.	NON-COPLANAR PARALLEL		Line of action of all forces not lying on the same plane but are parallel to each other.
9.	NON-COPLANAR NON-CONCURRENT NON-PARALLEL		Line of action of all forces not lying on same plane, which are not parallel to each other and also not passing through a same point.

Types of Parallel forces:

- LIKE PARALLEL FORCE → Parallel forces which are acting in the same direction are known as like parallel forces.
- UNLIKE PARALLEL FORCE → Parallel forces which are acting in the different direction are known as unlike parallel forces.



Like parallel force



Unlike parallel force.

Fundamental laws of mechanics:

- Newton's law of motion
- Newton's gravitational law
- Parallelogram law
- Principle of transmissibility

NEWTON'S FIRST LAW:

It states that, every particle continues in a state of rest or motion in a straight line unless it is acted upon by an external force.

NEWTON'S SECOND LAW:

It states that, 'if the resultant force acting on a particle is not zero, the acceleration of the particle is directly proportional to the force acting on it and inversely proportional to the mass of the particle.

$$(ie) a \propto \frac{F}{m} \Rightarrow F = ma$$

NEWTON'S THIRD LAW:

The law states that, for every action there is equal and opposite reaction.

NEWTON'S LAW OF GRAVITATION:

The law states that, a particle attracts with other particle in the universe with a force which is directly proportional to their masses and inversely proportional to the square of the distance between the centres.

$$(ie) F \propto \frac{m_1 m_2}{r^2} \Rightarrow F = G \cdot \frac{m_1 m_2}{r^2}$$

RESULTANT OF FORCE:

The resultant of number of forces acting on a particle is a single equivalent force which produces the same effect as that of all given forces. It can be analysed using (a) Parallelogram law (b) Triangle law

(c) Polygon law (d) Resolution of force.

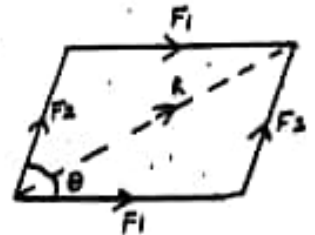
PARALLELOGRAM LAW:

If two coplanar con-current forces acting on a particle are represented by two adjacent sides of a parallelogram taken in order, then the diagonal connecting the two sides represents the resultant in opposite order.



Resultant Magnitude $R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$

Resultant Direction $\alpha = \tan^{-1} \left[\frac{F_2 \sin \theta}{F_1 + F_2 \cos \theta} \right]$

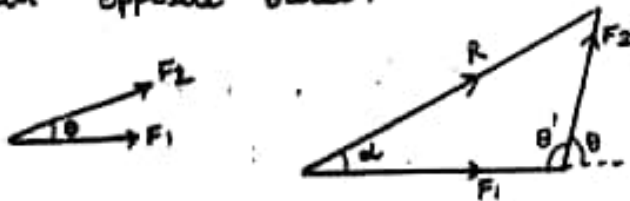


where, $\theta \rightarrow$ angle between two forces.

$\alpha \rightarrow$ angle which Resultant force makes with one of force

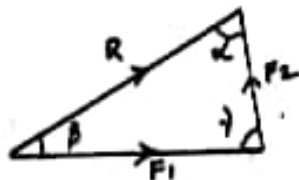
TRIANGLE LAW:

If two coplanar con-current forces acting on a particle are represented by two adjacent sides of a triangle taken in order, then the closing side represents the resultant in opposite order.



$$R = \sqrt{F_1^2 + F_2^2 - 2F_1F_2 \cos \theta'}$$

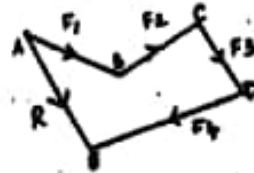
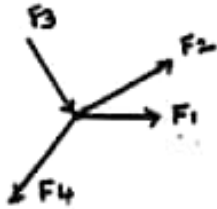
Using sine law,



$$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{R}{\sin \gamma}$$

POLYGON LAW:

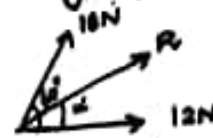
If number of co-planar concurrent forces acting simultaneously on a particle represented in magnitude and direction by the sides of open polygon taken in order, then the closing side of the polygon represents the resultant taken in opposite order.



PROBLEMS:

- ① Two concurrent forces of 12N and 18N are acting at an angle of 60° . Find the resultant force using parallelogram law.

Gn: $F_1 = 12\text{N}$; $F_2 = 18\text{N}$; $\theta = 60^\circ$.



Soln: Resultant Magnitude $R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$

$$= \sqrt{12^2 + 18^2 + (2 \times 12 \times 18 \times \cos 60)}$$

$$R = 26.15\text{N}$$

Resultant Direction $\alpha = \tan^{-1} \left[\frac{F_2 \sin \theta}{F_1 + F_2 \cos \theta} \right]$

$$= \tan^{-1} \left[\frac{18 \sin 60}{12 + 18 \cos 60} \right]$$
$$\alpha = 36.34^\circ$$

- ② Two concurrent forces act at an angle of 30° . Resultant force is 15N and one of the force is 10N. Find other force.

Gn: $F_1 = 10\text{N}$; $F_2 = ?$; $R = 15\text{N}$; $\theta = 30^\circ$.

Soln:

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$$
$$R^2 = F_1^2 + F_2^2 + 2F_1F_2\cos\theta$$

$$15^2 = 10^2 + F_2^2 + 2(10)F_2 \cos 30$$

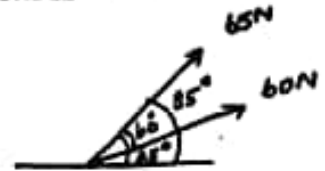
$$F_2^2 + 17.32 F_2 - 125 = 0$$

On solving, $F_2 = -22.8 \text{ N}$ or 5.48 N .

$$\therefore F_2 = 5.48 \text{ N}$$

- ③ Two forces 60 N and 65 N act on a screw at an angle of 25° and 85° from base. Determine the magnitude and direction of their resultant.

Given: $F_1 = 60 \text{ N}$; $F_2 = 65 \text{ N}$; $\theta = 85^\circ - 25^\circ = 60^\circ$



Soln: Resultant Magnitude $R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$

$$= \sqrt{60^2 + 65^2 + (2 \times 60 \times 65 \times \cos 60)}$$

$$R = 108.28 \text{ N}$$

Resultant Direction $\alpha = \tan^{-1} \left[\frac{F_2 \sin \theta}{F_1 + F_2 \cos \theta} \right]$

$$= \tan^{-1} \left[\frac{65 \sin 60}{60 + 65 \cos 60} \right]$$

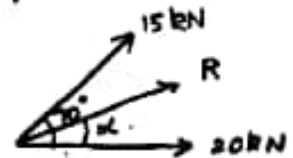
$$\alpha = 31.3^\circ$$

- ④ Two forces $F_1 = 20 \text{ kN}$ & $F_2 = 15 \text{ kN}$ acts on a particle as shown in figure. Determine resultant.

Resultant Magnitude $R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$

$$= \sqrt{20^2 + 15^2 + (2)(20)(15) \cos 70}$$

$$= 28.813 \text{ kN}$$



Resultant Direction $\alpha = \tan^{-1} \left[\frac{F_2 \sin \theta}{F_1 + F_2 \cos \theta} \right]$

$$= \tan^{-1} \left[\frac{15 \sin 70}{20 + 15 \cos 70} \right]$$

$$\alpha = 29.3^\circ$$

RESOLUTION OF FORCE:

The splitting up of a force into horizontal and vertical component along a fixed reference axis is called resolution of forces.

Horizontal Component:

From fig: $\Delta^{le} OAB$,

$$\cos \theta = \frac{F_x}{F}$$

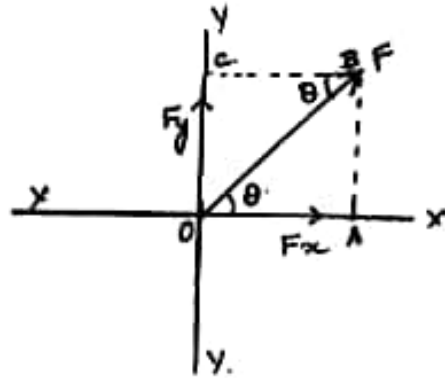
$$F_x = F \cos \theta$$

Vertical Component:

In $\Delta^{le} OBC$,

$$\sin \theta = \frac{F_y}{F}$$

$$F_y = F \sin \theta$$



Resultant Magnitude:

$$R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2}$$

Resultant Direction:

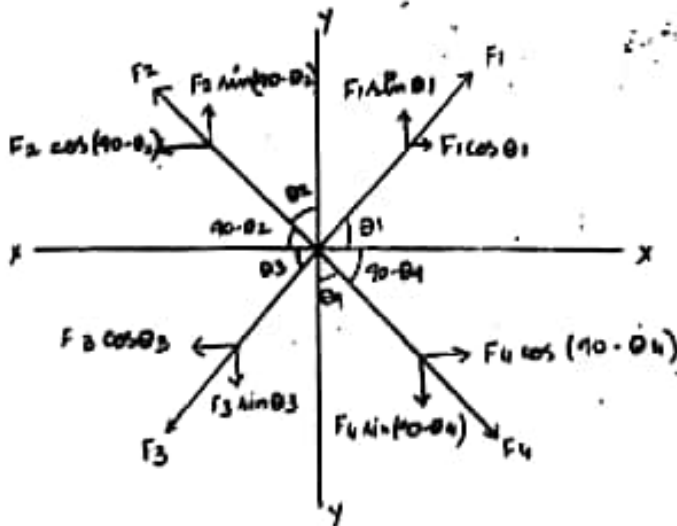
$$\alpha = \tan^{-1} \left| \frac{\sum F_y}{\sum F_x} \right|$$

where, $\sum F_x$ = Sum of all forces along x-direction.

$\sum F_y$ = Sum of all forces along y-direction.

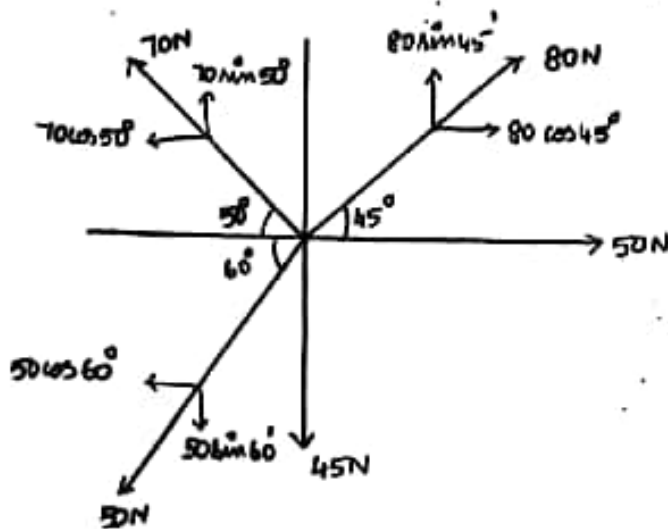
θ = angle between force and x-axis.

(i.e. angle w.r.t x-axis).



PROBLEMS:

① Find the resultant of the given force system.



Step 1: Sum of horizontal force:

$$\begin{aligned}\sum F_x &= 80 \cos 45^\circ - 70 \cos 50^\circ - 50 \cos 60^\circ + 50 \\ &= 36.57 \text{ N}\end{aligned}$$

Step 2: Sum of vertical force:

$$\begin{aligned}\sum F_y &= 80 \sin 45^\circ + 70 \sin 50^\circ - 50 \sin 60^\circ - 45 \\ &= 21.89 \text{ N}\end{aligned}$$

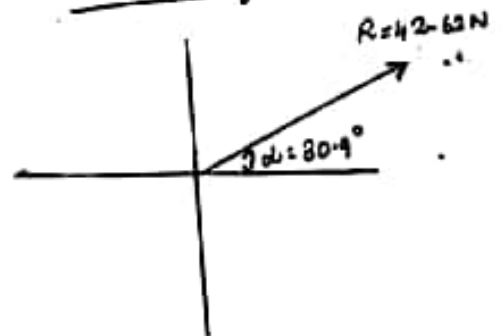
Step 3: Resultant Magnitude:

$$R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2} = \sqrt{(36.57)^2 + (21.89)^2} = 42.62 \text{ N}$$

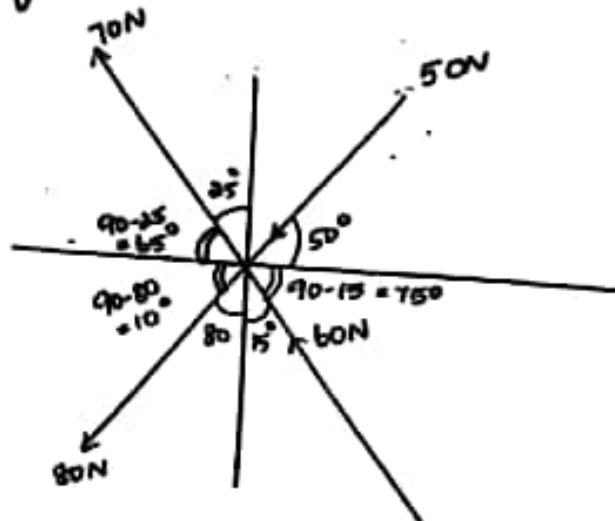
Step 4: Resultant Direction:

$$\begin{aligned}\alpha &= \tan^{-1} \left| \frac{\sum F_y}{\sum F_x} \right| \\ &= \tan^{-1} \left| \frac{21.89}{36.57} \right| = 30.9^\circ\end{aligned}$$

Location of Resultant:



② Find the resultant magnitude and direction for the given system of forces:



Step 1: Sum of horizontal force:

$$\begin{aligned}\sum F_x &= -50 \cos 50^\circ - 70 \cos 65^\circ - 80 \cos 10^\circ - 60 \cos 75^\circ \\ &= -156.031 \text{ N}\end{aligned}$$

Step 2: Sum of Vertical force:

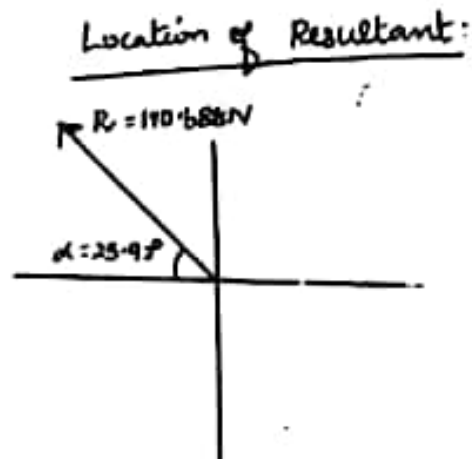
$$\begin{aligned}\sum F_y &= -50 \sin 50^\circ + 70 \sin 65^\circ - 80 \sin 10^\circ + 60 \sin 75^\circ \\ &= 69.203 \text{ N}.\end{aligned}$$

Step 3: Resultant Magnitude:

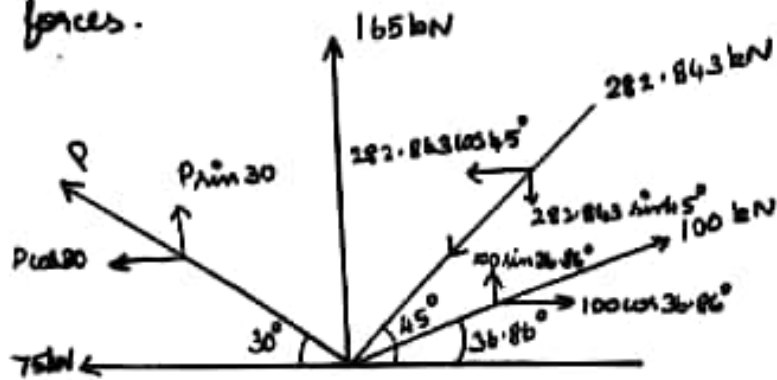
$$\begin{aligned}R &= \sqrt{\sum F_x^2 + \sum F_y^2} \\ &= \sqrt{(-156.03)^2 + (69.203)^2} = 170.688 \text{ N}\end{aligned}$$

Step 4: Resultant Direction:

$$\begin{aligned}\alpha &= \tan^{-1} \left| \frac{\sum F_y}{\sum F_x} \right| \\ &= \tan^{-1} \left| \frac{69.203}{-156.03} \right| \\ \alpha &= 23.92^\circ\end{aligned}$$



- ③ If 5 forces acting on a particle as shown in figure & the algebraic sum of all the horizontal force is -324.904 kN . Calculate the force magnitude (P) & the resultant of all the forces.



$$\Sigma F_x = -75 - P \cos 30^\circ - 282.843 \cos 45^\circ + 100 \cos 36.86^\circ$$

Given that $\Sigma F_x = -324.904 \text{ kN}$

$$\therefore -324.904 = -194.99 - P \cos 30^\circ$$

$$P \cos 30^\circ = 129.914$$

$$\boxed{P = 150.01 \text{ kN}}$$

$$\Sigma F_y = P \sin 30^\circ + 165 - 282.843 \sin 45^\circ + 100 \sin 36.86^\circ$$

$$= 100 \text{ kN}$$

Resultant $R = \sqrt{\Sigma F_x^2 + \Sigma F_y^2}$

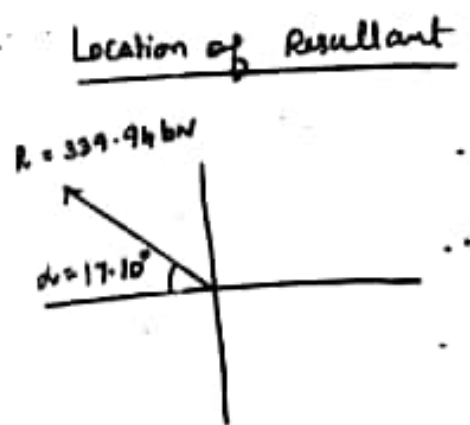
$$= \sqrt{(-324.904)^2 + (100)^2}$$

$$R = 339.94 \text{ kN}$$

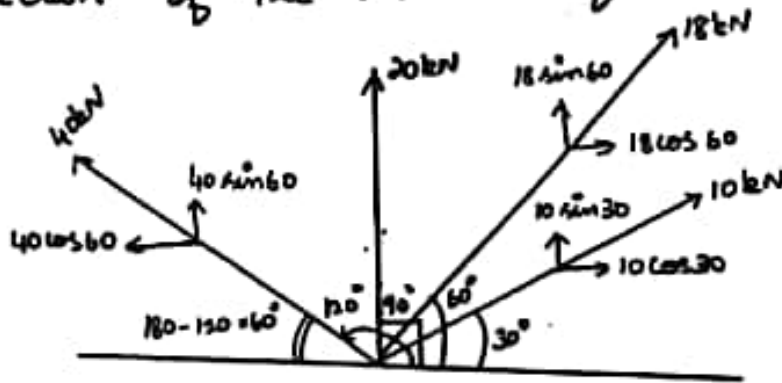
$$\alpha = \tan^{-1} \left| \frac{\Sigma F_y}{\Sigma F_x} \right|$$

$$= \tan^{-1} \left| \frac{100}{-324.904} \right|$$

$$\alpha = 17.10^\circ$$



- ④ Four co-planar concurrent forces of magnitude 10kN, 18kN, 20kN and 40kN are acting on a particle. The angle made by the above forces with x-axis are 30° , 60° , 90° and 120° respectively. Find the magnitude and direction of the resultant force.



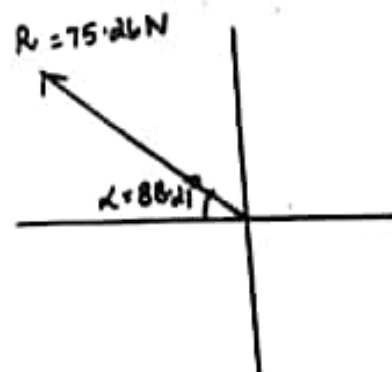
$$\begin{aligned}\sum F_x &= 10 \cos 30^\circ + 18 \cos 60^\circ - 40 \cos 60^\circ \\ &= -2.34 \text{ kN}\end{aligned}$$

$$\begin{aligned}\sum F_y &= 10 \sin 30^\circ + 18 \sin 60^\circ + 20 + 40 \sin 60^\circ \\ &= 75.23 \text{ kN}\end{aligned}$$

$$\begin{aligned}\text{Resultant } R &= \sqrt{\sum F_x^2 + \sum F_y^2} \\ &= \sqrt{(-2.34)^2 + (75.23)^2} \\ &= 75.26 \text{ kN}\end{aligned}$$

Location of Resultant

$$\begin{aligned}\alpha &= \tan^{-1} \left| \frac{\sum F_y}{\sum F_x} \right| \\ &= \tan^{-1} \left| \frac{75.23}{-2.34} \right| \\ \alpha &= 88.21^\circ\end{aligned}$$



- ⑤ Four forces of magnitude 10N, 20N, 30N and 40N are acting along the 4 sides of the square ABCD as shown in figure. Determine the magnitude and direction of given force

$$\sum F_x = 10 - 30 = -20\text{N}$$

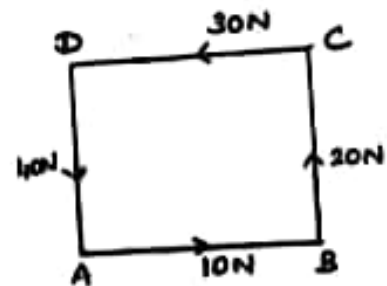
$$\sum F_y = 20 - 40 = -20\text{N}$$

$$R = \sqrt{\sum F_x^2 + \sum F_y^2}$$

$$= \sqrt{(-20)^2 + (-20)^2} = 28.28\text{N}$$

$$\alpha = \tan^{-1} \left| \frac{\sum F_y}{\sum F_x} \right|$$

$$= \tan^{-1} \left| \frac{-20}{-20} \right| \Rightarrow \alpha = 45^\circ$$



- ⑥ A triangle ABC has its sides AB = 4m, BC = 3m. Three forces of 40N, 50N and 30N are acting along the sides AB, BC and CA respectively. Determine the resultant force

To find θ :

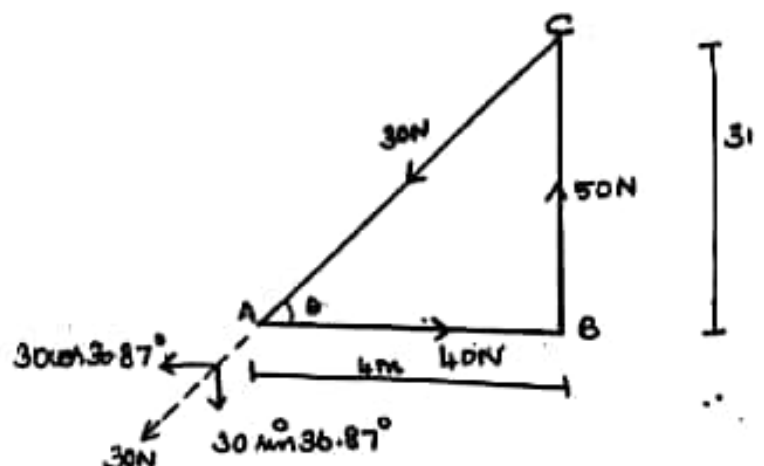
$$AC^2 = AB^2 + BC^2$$

$$AC = \sqrt{3^2 + 4^2} = 5\text{m}$$

$$\tan \theta = \frac{3}{4}$$

$$\theta = \tan^{-1}(3/4)$$

$$\theta = 36.87^\circ$$



$$\Sigma F_x = 40 - 30 \cos 36.87^\circ$$

$$= 15.997 \text{ N}$$

$$\Sigma F_y = 50 - 30 \sin 36.87^\circ$$

$$= 32.004 \text{ N}$$

$$\text{Resultant } R = \sqrt{\Sigma F_x^2 + \Sigma F_y^2}$$

$$= \sqrt{(15.997)^2 + (32.004)^2}$$

$$R = 35.78 \text{ N}$$

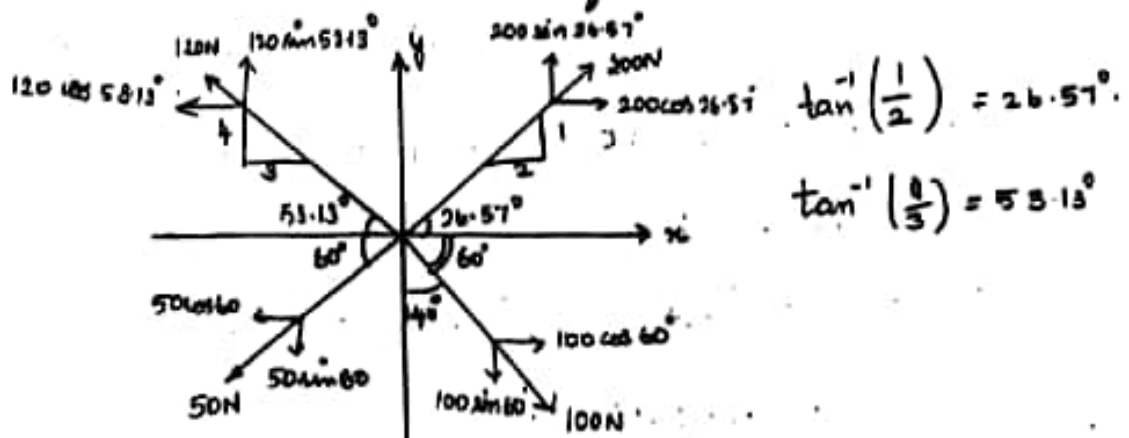
$$\alpha = \tan^{-1} \left| \frac{\Sigma F_y}{\Sigma F_x} \right|$$

$$= \tan^{-1} \left| \frac{32.004}{15.997} \right|$$

$$\alpha = 68.44^\circ$$

(P0)

7 A system of four forces acting on a body is shown in figure. Determine the resultant force and its direction.



$$\Sigma F_x = 200 \cos 26.57^\circ - 120 \cos 53.13^\circ - 50 \cos 60^\circ + 100 \cos 60^\circ$$

$$= 131.88 \text{ N}$$

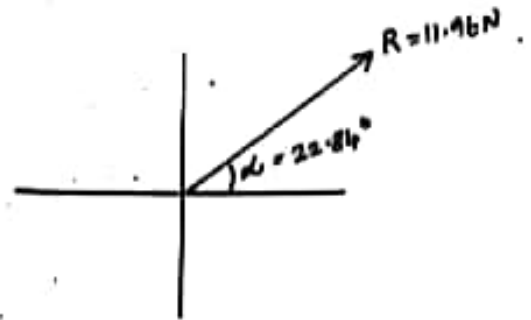
$$\Sigma F_y = 200 \sin 26.57^\circ + 120 \sin 53.13^\circ - 50 \sin 60^\circ - 100 \sin 60^\circ$$

$$= 55.55 \text{ N}$$

$$\begin{aligned}\text{Resultant Magnitude } R &= \sqrt{\sum F_x^2 + \sum F_y^2} \\ &= \sqrt{(131.88)^2 + (55.55)^2} \\ &= 11.96 \text{ N}\end{aligned}$$

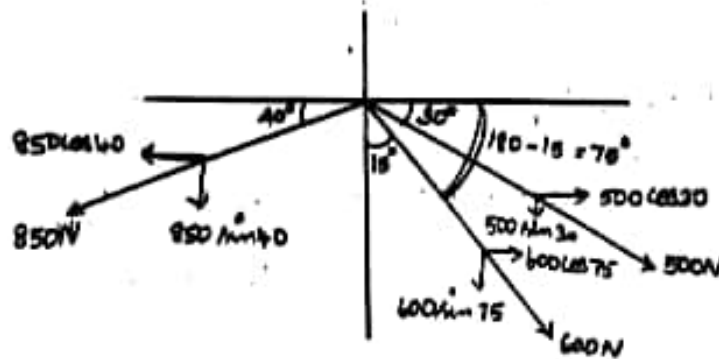
$$\begin{aligned}\text{Direction } \alpha &= \tan^{-1} \left| \frac{\sum F_y}{\sum F_x} \right| \\ &= \tan^{-1} \left| \frac{55.55}{131.88} \right|\end{aligned}$$

$$\alpha = 22.84^\circ$$



pu
8

Determine the magnitude and direction of the resultant for the forces acting on a particle as shown in fig:



$$\begin{aligned}\sum F_x &= -850 \cos 40 + 600 \cos 75 + 500 \cos 30 \\ &= -62.83 \text{ N}\end{aligned}$$

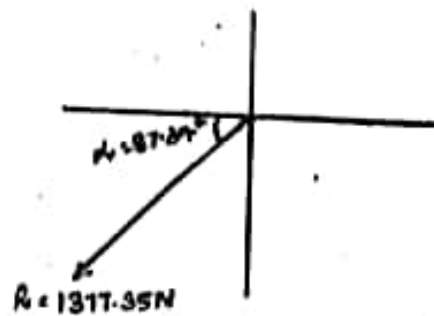
$$\begin{aligned}\sum F_y &= -850 \sin 40 - 600 \sin 75 - 500 \sin 30 \\ &= -1375.92 \text{ N}\end{aligned}$$

$$\begin{aligned}\text{Resultant } R &= \sqrt{\sum F_x^2 + \sum F_y^2} \\ &= \sqrt{(-62.83)^2 + (-1375.92)^2} \\ R &= 1377.35 \text{ N}\end{aligned}$$

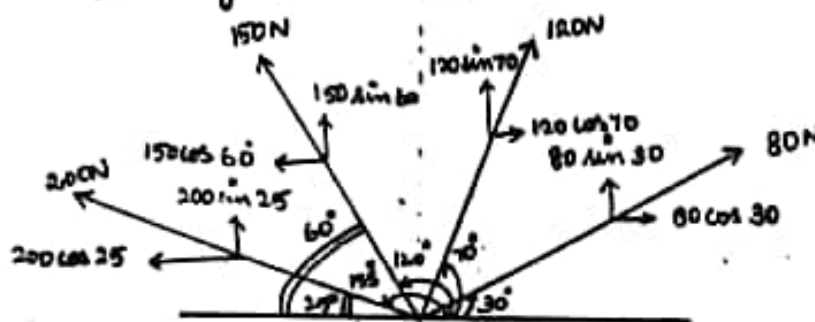
$$\alpha = \tan^{-1} \left| \frac{\sum F_y}{\sum F_x} \right|$$

$$= \tan^{-1} \left| \frac{-1375.92}{-62.83} \right|$$

$$\alpha = 87.39^\circ$$



- 9) Find the magnitude and direction of the concurrent forces of 80N, 120N, 150N and 200N making angles 30° , 70° , 120° and 155° respectively with x-axis.



$$\sum F_x = 80 \cos 30 + 120 \cos 70 - 150 \cos 60 - 200 \cos 25$$

$$= -145.94 \text{ N}$$

$$\sum F_y = 80 \sin 30 + 120 \sin 70 + 150 \sin 60 + 200 \sin 25$$

$$= 367.19 \text{ N}$$

Resultant Magnitude $R = \sqrt{\sum F_x^2 + \sum F_y^2}$

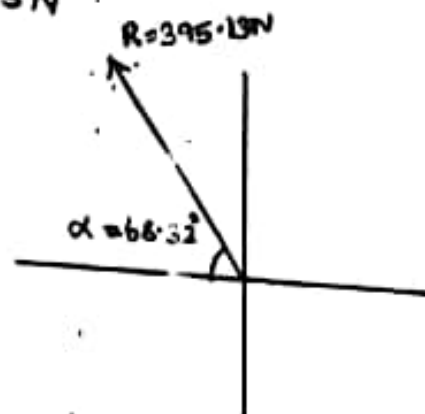
$$= \sqrt{(-145.94)^2 + (367.19)^2}$$

$$R = 395.13 \text{ N}$$

Resultant Direction $\alpha = \tan^{-1} \left| \frac{\sum F_y}{\sum F_x} \right|$

$$= \tan^{-1} \left| \frac{367.19}{-145.94} \right|$$

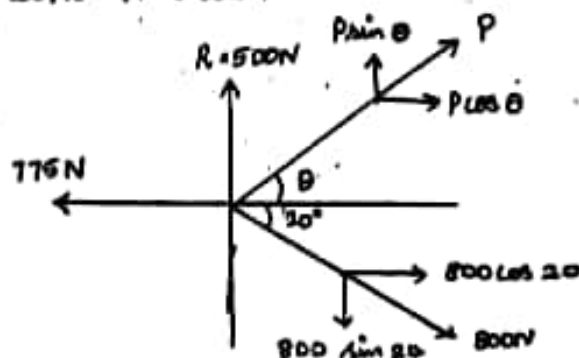
$$\alpha = 68.32^\circ$$



(90)

(10)

The three co-planar forces are acting at a point as shown in figure. One of the forces is unknown and its magnitude is shown by P . The resultant is having a magnitude 500N and is acting along y -axis (Positive direction). Determine the unknown force P and its inclination with x -axis.



As the resultant is vertical, it is equal to net vertical force and the net horizontal force is zero.

$$(ii) \quad \sum F_y = 500\text{N} \quad ; \quad \sum F_x = 0$$

$$\sum F_x = P \cos \theta - 775 + 800 \cos 20.$$

$$0 = P \cos \theta - 23.25$$

$$P \cos \theta = 23.25 \quad \text{--- (1)}$$

$$\sum F_y = P \sin \theta - 800 \sin 20.$$

$$500 = P \sin \theta - 173.62$$

$$P \sin \theta = 673.62 \quad \text{--- (2)}$$

$$\frac{(2)}{(1)} \Rightarrow \frac{P \sin \theta}{P \cos \theta} = \frac{673.62}{23.25}$$

$$\tan \theta = 28.97$$

$$\theta = \tan^{-1}(28.97)$$

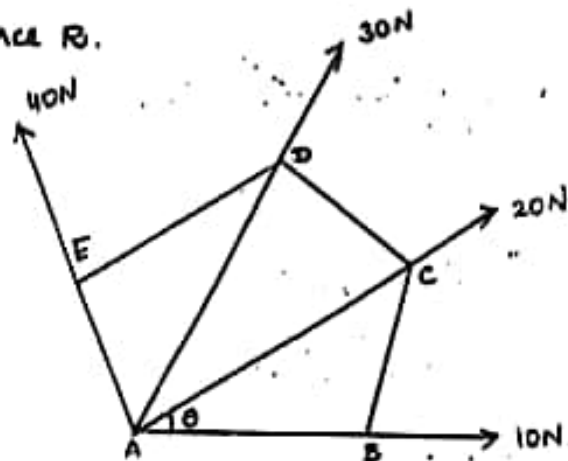
$$\theta = 88.27^\circ$$

Sub-1 $\theta = 88.27^\circ$ in ①,

① $\rightarrow P \cos 88.27^\circ = 23.25$

$$P = \frac{23.25}{\cos 88.27^\circ} \Rightarrow \boxed{P = 770.13 \text{ N}}$$

- ② The forces 10N, 20N, 30N and 40N are acting on one of the vertices of a regular pentagon, towards the other four vertices taken in order. Find the magnitude and direction of the resultant force R.



Sum of interior angles of regular polygon $= (2n-4) \times 90^\circ$.

where n = no. of sides. So, for regular pentagon,

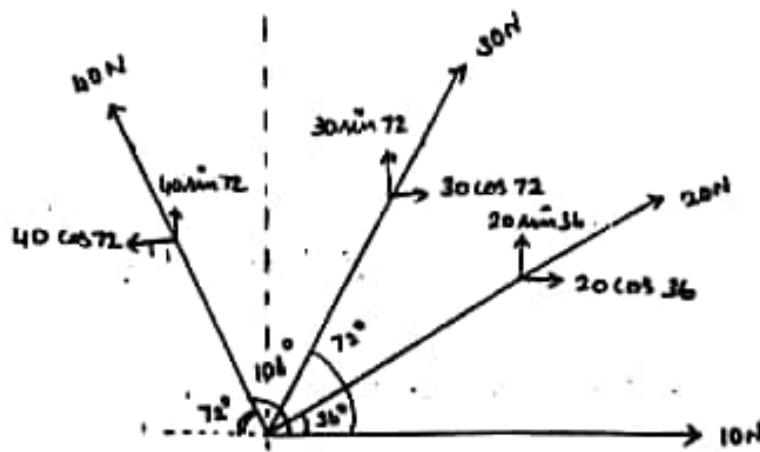
$$\text{Sum of interior angles} = (2 \times 5 - 4) \times 90 = 540^\circ.$$

$$\therefore \text{Each included angle} = \frac{540^\circ}{5} = 108^\circ.$$

$$\therefore \text{Angle } \theta = \frac{108^\circ}{3} = 36^\circ.$$

Hence,

$F_1 = 10\text{N}$	$\theta_1 = 0^\circ$
$F_2 = 20\text{N}$	$\theta_2 = 36^\circ$
$F_3 = 30\text{N}$	$\theta_3 = 2 \times 36 = 72^\circ$
$F_4 = 40\text{N}$	$\theta_4 = 3 \times 36 = 108^\circ$



$$\begin{aligned}\Sigma F_x &= 10 + 20 \cos 36^\circ + 30 \cos 72^\circ - 40 \cos 72^\circ \\ &= 23.09 \text{ N}\end{aligned}$$

$$\begin{aligned}\Sigma F_y &= 20 \sin 36^\circ + 30 \sin 72^\circ + 40 \sin 72^\circ \\ &= 78.33 \text{ N}\end{aligned}$$

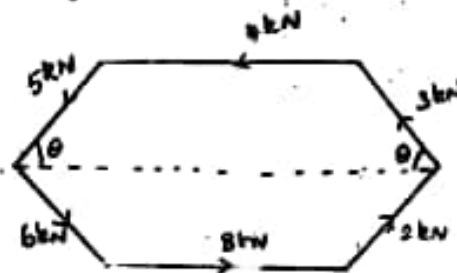
$$\begin{aligned}\text{Resultant } R &= \sqrt{\Sigma F_x^2 + \Sigma F_y^2} \\ &= \sqrt{(23.09)^2 + (78.33)^2}\end{aligned}$$

$$R = 81.67 \text{ N}$$

$$\alpha = \tan^{-1} \left| \frac{\Sigma F_y}{\Sigma F_x} \right|$$

$$= \tan^{-1} \left| \frac{78.33}{23.09} \right| \Rightarrow \alpha = 73.57^\circ$$

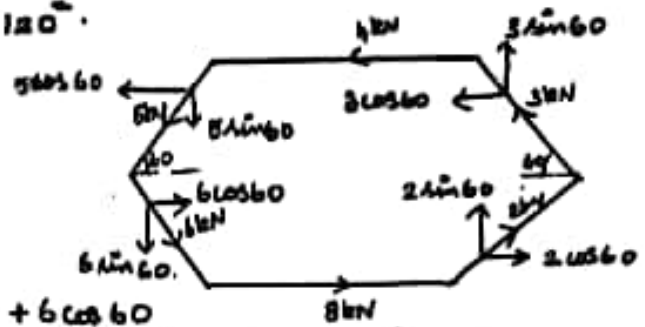
- ⑫ Six forces acting on a regular hexagon are 8kN, 2kN, 3kN, 4kN, 5kN, 6kN. Find the magnitude and direction of the resultant force.



Sum of interior angle in regular hexagon = $(2 \times 6 - 4) \times 90 = 720^\circ$.

$$\therefore \text{each angle} = \frac{720^\circ}{6} = 120^\circ.$$

$$\text{Angle } \theta = \frac{120^\circ}{2} = 60^\circ.$$



$$\Sigma F_x = 8 + 2 \cos 60 - 3 \cos 60 - 4 - 5 \cos 60 + 6 \cos 60$$

$$= 4 \text{ kN}$$

$$\Sigma F_y = 2 \sin 60 + 3 \sin 60 - 5 \sin 60 - 6 \sin 60$$

$$= -5.20 \text{ kN}$$

$$R = \sqrt{\Sigma F_x^2 + \Sigma F_y^2}$$

$$= \sqrt{(4)^2 + (-5.2)^2} \Rightarrow R = 6.56 \text{ kN}$$

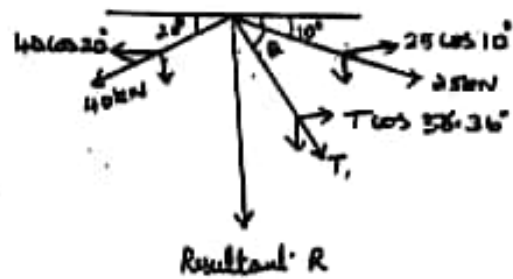
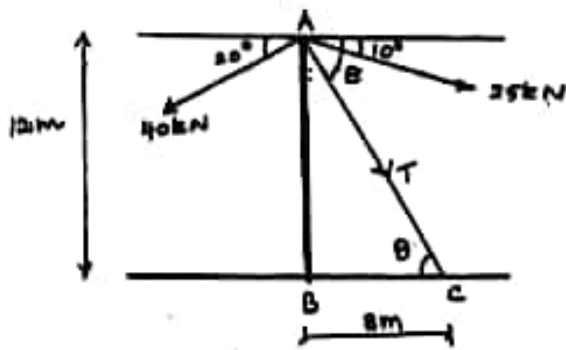
$$\alpha = \tan^{-1} \left| \frac{\Sigma F_y}{\Sigma F_x} \right|$$

$$= \tan^{-1} \left| \frac{-5.2}{4} \right| \Rightarrow \alpha = 52.43^\circ$$

- (15) Two cables which have known tensions are attached to the top of a tower AB. A third cable AC is used as a guy wire as shown in figure. Determine the tension in AC if the resultant of the forces exerted at A by the three cables acts vertically downward.

It is given that the resultant force is acting vertically downward. Hence, the net horizontal force is zero and the net vertical force is equal to resultant force.

$$(a) \quad \Sigma F_x = 0 \quad ; \quad \Sigma F_y = R.$$



In right angled triangle, ABC,

$$\tan \theta = \frac{12}{8} \Rightarrow \theta = 56.30^\circ$$

Resolving the forces horizontally,

$$\Sigma F_x = 25 \cos 10^\circ + T \cos 56.30^\circ - 40 \cos 20^\circ$$

$$0 = 24.62 + 0.555T - 37.58$$

$$T = 23.35 \text{ N}$$

$\therefore$ Tension in the Guy wire is 23.35 N.

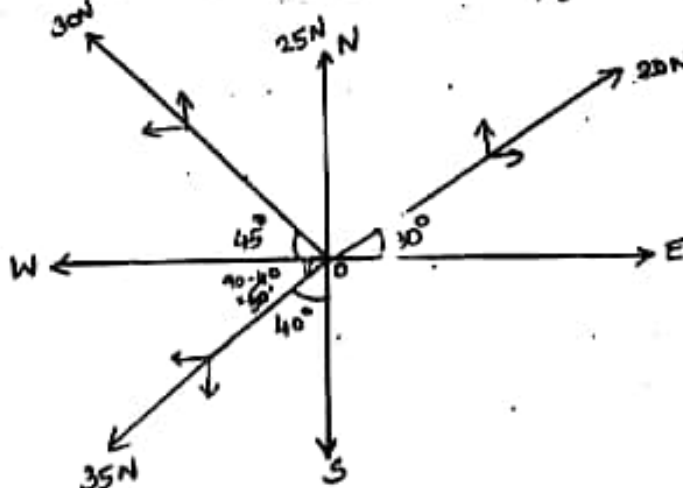
(14) Particle O is acted on by following forces. Find the resultant.

a) 20N inclined 30° to north of east.

b) 25N towards the north.

c) 30N towards north-west

d) 35N inclined 40° to west of south.



$$\Sigma F_x = 20 \cos 30 - 30 \cos 45 - 35 \cos 50$$

$$= -26.39 \text{ N}$$

$$\Sigma F_y = 20 \sin 30 + 25 + 30 \sin 45 - 35 \sin 50$$

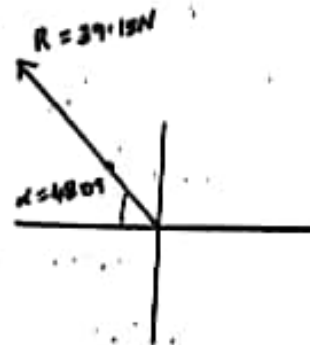
$$= 29.40 \text{ N}$$

$$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$$

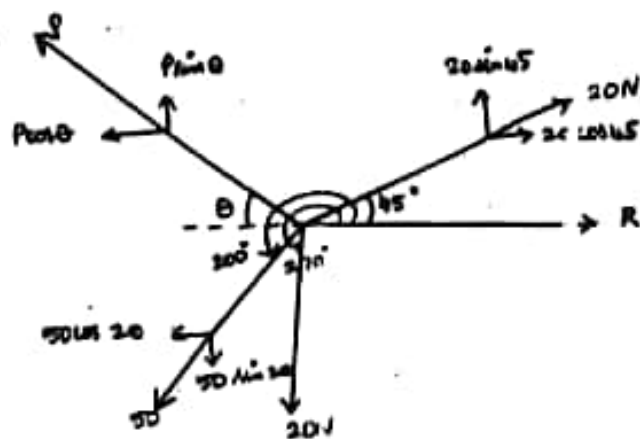
$$= \sqrt{(-26.39)^2 + (29.40)^2} = 39.51 \text{ N}$$

$$\alpha = \tan^{-1} \left| \frac{\Sigma F_y}{\Sigma F_x} \right|$$

$$= \tan^{-1} \left| \frac{29.40}{-26.39} \right| = 48.09^\circ$$



- (15) Four co-planar forces are acting at a point as shown in figure. Three forces have magnitude of 20N, 50N & 20N at an angle of 45° , 200° & 270° respectively. Fourth force is unknown. Determine the unknown force & its direction from x-axis if the resultant force has magnitude 50N acting positive direction.



$$\Sigma F_x = 10 \cos 45^\circ - P \cos \theta - 50 \cos 20^\circ$$

On that $\Sigma F_x = 50N$

$$50 = 11.14 - P \cos \theta - 46.98$$

$$-P \cos \theta = 46.98 - 11.14 + 50$$

$$P \cos \theta = -82.84 \quad \text{--- (1)}$$

$$\Sigma F_y = 20 \sin 45^\circ + P \sin \theta - 50 \sin 20^\circ - 20$$

On that $\Sigma F_y = 0$

$$0 = 11.14 + P \sin \theta - 17.10 - 20$$

$$P \sin \theta = 22.95 \quad \text{--- (2)}$$

$$\frac{(2)}{(1)} \Rightarrow \frac{P \sin \theta}{P \cos \theta} = \frac{22.95}{-82.84}$$

$$\tan \theta = (-0.27)$$

$$\theta = 15.14^\circ$$

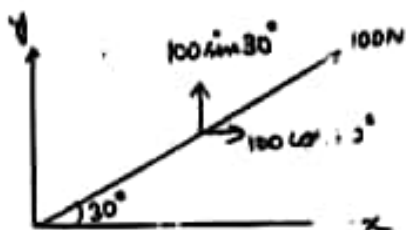
Sub θ in (1) $\Rightarrow P \cos 15.14^\circ = -82.84$

$$P = -85.82N$$

(9)

(16)

A force of magnitude 100N in x-y plane is inclined at 30° to x-direction. Determine the components of force along x & y direction.



$$F_x = 100 \cos 30^\circ = 86.6N$$

$$F_y = 100 \sin 30^\circ = 50N$$

EQUILIBRIUM OF PARTICLES:

A body is said to be in equilibrium when the resultant force of the system of force acting on it is zero.

EQUATION OF EQUILIBRIUM:

For equilibrium condition, resultant is zero.

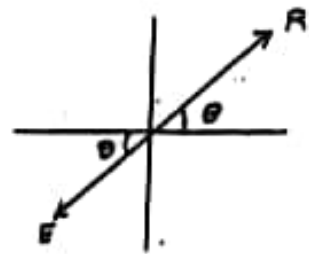
$$(i) R = 0$$

$$R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2} = 0$$

$$\Rightarrow \sum F_x = 0 \quad \text{and} \quad \sum F_y = 0$$

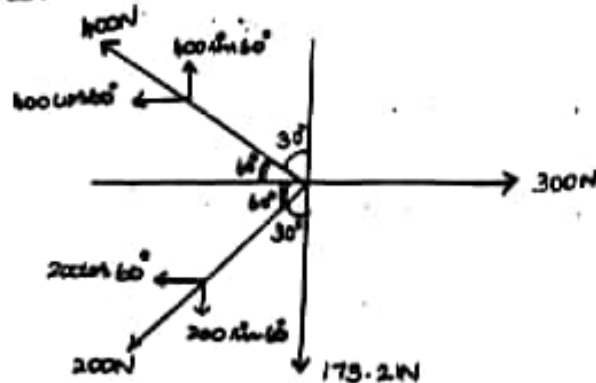
EQUILIBRANT (E):

It is a force which is equal in magnitude and opposite in direction to the resultant.



PROBLEMS:

- ① Examine the equilibrium condition for the particle shown in figure.



For equilibrium condition,

$$\sum F_x = 0 \quad \text{and} \quad \sum F_y = 0$$

$$\sum F_x = 300 - 400 \cos 60^\circ - 200 \cos 60^\circ$$

$$= 0$$

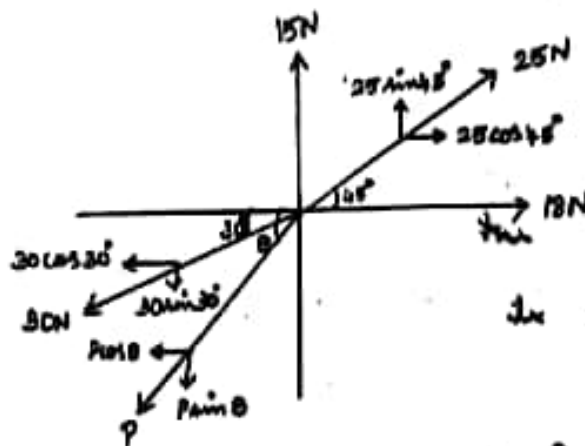
$$\Sigma F_y = 400 \sin 60^\circ - 200 \sin 60^\circ - 173.21$$

$$= 0$$

$$\therefore R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$$

$$= \sqrt{0^2 + 0^2} \Rightarrow \boxed{R = 0}$$

- ② The forces shown in figure are in equilibrium. Find the magnitude and direction of unknown force P.



Since the forces are in equilibrium,

$$\Sigma F_x = 0 \quad \text{and} \quad \Sigma F_y = 0$$

$$\therefore \Sigma F_x = 18 + 25 \cos 45^\circ - 30 \cos 30^\circ + P \cos \theta = 0$$

$$P \cos \theta = 9.7 \quad \text{--- (1)}$$

$$\Sigma F_y = 25 \sin 45^\circ + 15 - 30 \sin 30^\circ - P \sin \theta = 0$$

$$P \sin \theta = 17.68 \quad \text{--- (2)}$$

$$\frac{(2)}{(1)} \Rightarrow \frac{P \sin \theta}{P \cos \theta} = \frac{17.68}{9.7}$$

$$\tan \theta = 1.82$$

$$\theta = \tan^{-1}(1.82)$$

$$\boxed{\theta = 61.24^\circ}$$

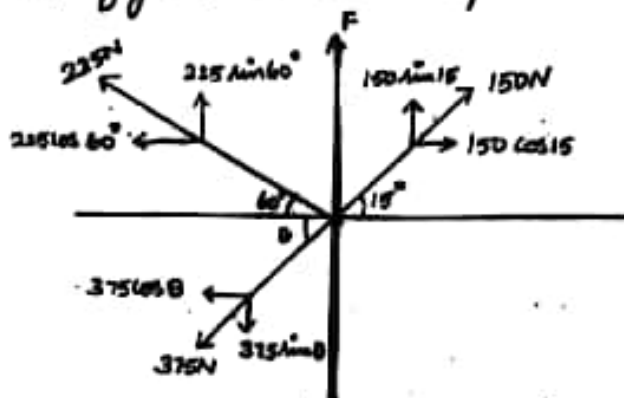
Sub-1 $\theta = 61.24^\circ$ in ①,

$$\textcircled{1} \Rightarrow P \cos 61.24^\circ = 9.7$$

$$P = \frac{9.7}{\cos 61.24^\circ}$$

$$\boxed{P = 20.16 \text{ N}}$$

③ Determine the magnitude of force F and θ for the particle shown in figure which is in equilibrium.



Since the particle is in equilibrium,

$$\sum F_x = 0 \quad \text{and} \quad \sum F_y = 0$$

$$\sum F_x = 150 \cos 15^\circ - 225 \cos 60^\circ - 375 \cos \theta = 0$$

$$144.88 - 112.5 - 375 \cos \theta = 0$$

$$375 \cos \theta = 32.38$$

$$\cos \theta = \frac{32.38}{375} = 0.086$$

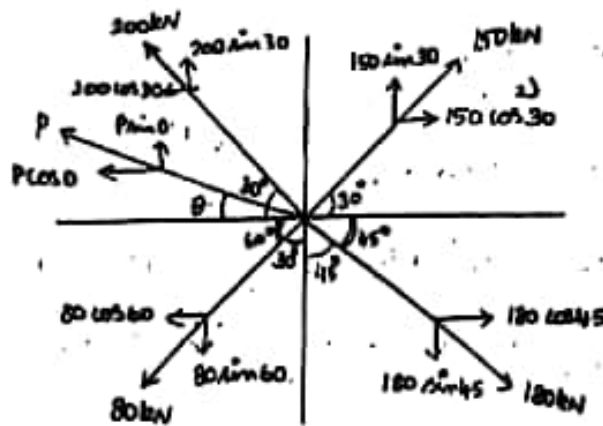
$$\theta = \cos^{-1}(0.086) \Rightarrow \boxed{\theta = 85.05^\circ}$$

$$\sum F_y = 150 \sin 15^\circ + 225 \sin 60^\circ + F - 375 \sin \theta = 0$$

$$150 \sin 15^\circ + 225 \sin 60^\circ + F - 375 \sin 85.05^\circ = 0$$

On solving, $\boxed{F = 139.92 \text{ N}}$

- ④ A system of co-planar concurrent forces as shown in figure are in equilibrium. Find the magnitude of P and θ .



Since the system is in equilibrium

$$\sum F_x = 0 \quad \text{and} \quad \sum F_y = 0$$

$$\sum F_x = 150 \cos 30^\circ - 200 \cos 30^\circ - P \cos \theta - 80 \cos 60^\circ + 180 \cos 45^\circ = 0$$

$$P \cos \theta = 43.98 \quad \text{--- (1)}$$

$$\sum F_y = 150 \sin 30^\circ + 200 \sin 30^\circ - 80 \sin 60^\circ - 180 \sin 45^\circ + P \sin \theta = 0$$

$$P \sin \theta = 21.55 \quad \text{--- (2)}$$

$$\frac{(2)}{(1)} \Rightarrow \frac{P \sin \theta}{P \cos \theta} = \frac{21.55}{43.98}$$

$$\tan \theta = 0.49$$

$$\theta = \tan^{-1}(0.49) \Rightarrow \boxed{\theta = 26.10^\circ}$$

Sub- θ in (1),

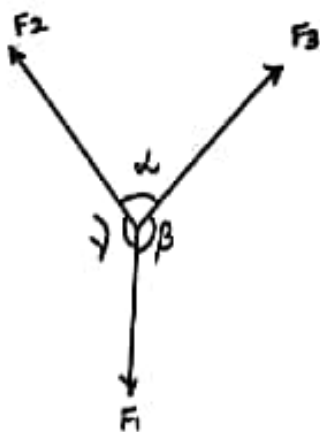
$$P \cos 26.10^\circ = 43.98$$

$$P = \frac{43.98}{\cos 26.10^\circ}$$

$$\boxed{P = 48.97 \text{ kN}}$$

LAMI'S THEOREM:

It states that if a particle is in equilibrium under the action of three forces, then each force is proportional to the sine of angle between the other two forces.

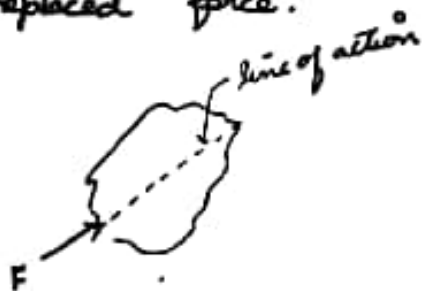


Mathematically,

$$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$$

PRINCIPLE OF TRANSMISSIBILITY:

The state of rest or motion of a body is unchanged if a force acting on a body is replaced by another force of same magnitude and direction but acting anywhere on the body along the line of action of the replaced force.



or



FREE BODY DIAGRAM:

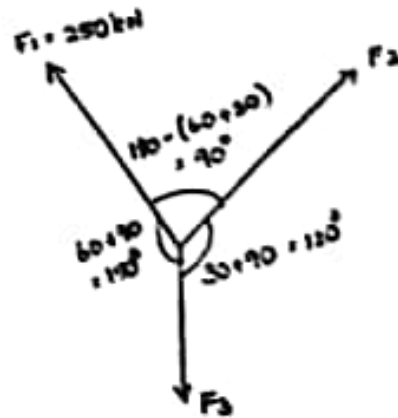
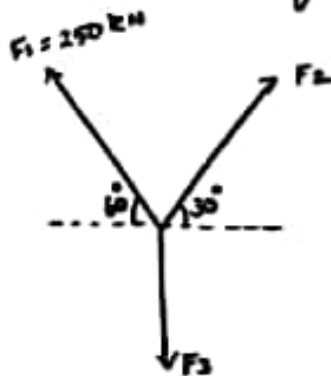
It is a diagram of a body obtained by removing the contact surface and replacing it by a reaction force perpendicular to the surface.

S.No	ACTUAL DIAGRAM	FREE BODY DIAGRAM (FBD)
1		
2		
3		
4		
5		
6		
7		

S.No	ACTUAL DIAGRAM	FREE BODY DIAGRAM (FBD)
8		<u>FBD - A</u> <u>FBD - B</u>
9		
10		<u>FBD - A</u> <u>FBD - B</u>
11		<u>FBD - A</u> <u>FBD - B</u>
12		<u>FBD - A</u> <u>FBD - B</u>
13		<u>FBD - A</u> <u>FBD - B</u> <u>FBD - C</u>

PROBLEMS:

- ① The forces shown in figure are acting on a particle and in equilibrium. The magnitude of force $F_1 = 250 \text{ kN}$. Find F_2 & F_3 .



Applying Lami's theorem,

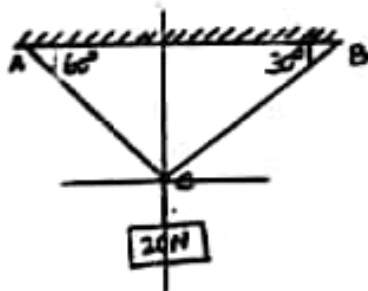
$$\frac{F_1 = 250}{\sin 120^\circ} = \frac{F_2}{\sin 150^\circ} = \frac{F_3}{\sin 90^\circ}$$

On Equating, $\frac{F_2}{\sin 150^\circ} = \frac{250}{\sin 120^\circ} \Rightarrow F_2 = 144.33 \text{ kN}$

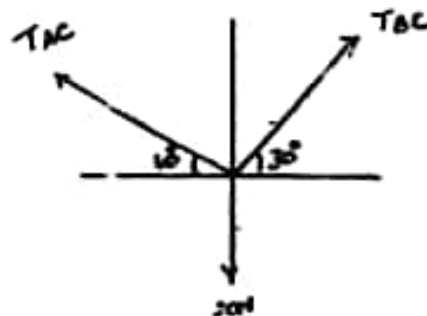
On Equating, $\frac{F_3}{\sin 90^\circ} = \frac{250}{\sin 120^\circ} \Rightarrow F_3 = 288.67 \text{ N}$

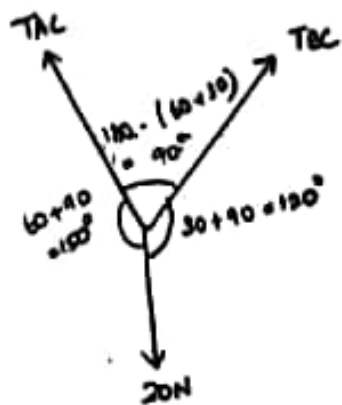
Q2

- ② A weight of 20 N is supported by two wires as shown in figure. Find the forces in the wire using Lami's theorem.



Free Body Diagram





Applying Lami's theorem,

$$\frac{20}{\sin 90^\circ} = \frac{T_{AC}}{\sin 120^\circ} = \frac{T_{BC}}{\sin 150^\circ}$$

On equating, $\frac{T_{AC}}{\sin 120^\circ} = \frac{20}{\sin 90^\circ}$

$$T_{AC} = \frac{20}{\sin 90^\circ} \times \sin 120^\circ$$

$$T_{AC} = 17.32 \text{ N}$$

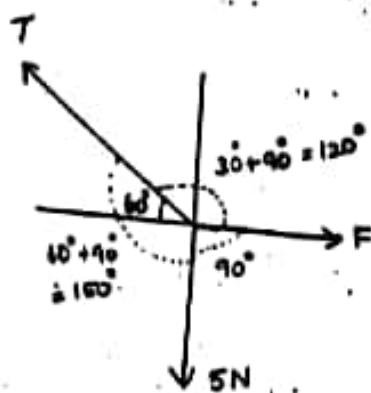
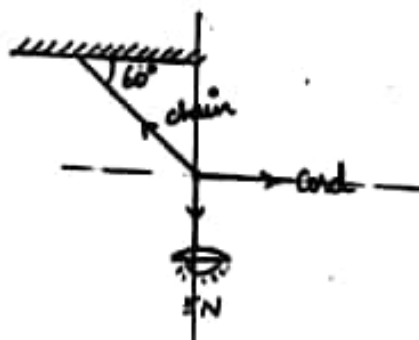
On equating, $\frac{T_{BC}}{\sin 150^\circ} = \frac{20}{\sin 90^\circ}$

$$T_{BC} = \frac{20}{\sin 90^\circ} \times \sin 150^\circ$$

$$T_{BC} = 10 \text{ N}$$

(PU)

- ③ A lamp weighing 5N is suspended from ceiling by a chain. It is pulled aside by a horizontal cord until the chain makes an angle of 60° with ceiling as shown in figure. Find the tensions in the chain and the cord.



Applying Lami's theorem,

$$\frac{5}{\sin 120^\circ} = \frac{F}{\sin 150^\circ} = \frac{T}{\sin 90^\circ}$$

On equating, $\frac{F}{\sin 150} = \frac{5}{\sin 120}$

$F = 2.89 \text{ N}$

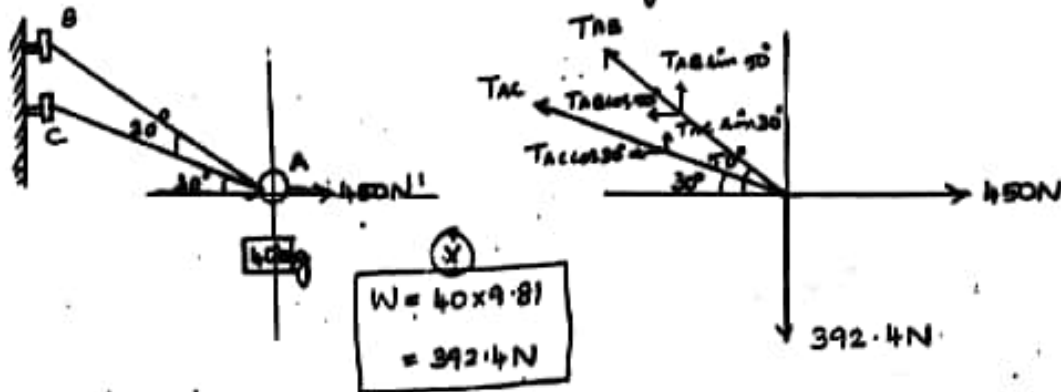
On equating, $\frac{T}{\sin 90} = \frac{5}{\sin 120}$

$T = 5.77 \text{ N}$

PU

4

Determine the tension in cables AB and AC required to hold the 40 kg crate as shown in figure.
 ↳ Convert into weight (N).



Applying resolution of forces,

$\sum F_x = 0$

$450 - T_{AB} \cos 50^\circ - T_{AC} \cos 30^\circ = 0$

$T_{AB} \cos 50^\circ + T_{AC} \cos 30^\circ = 450 \text{ N} \quad \text{--- (1)}$

$\sum F_y = 0$

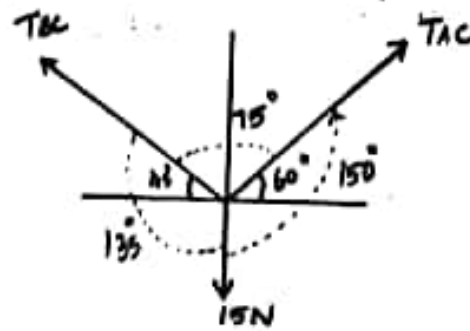
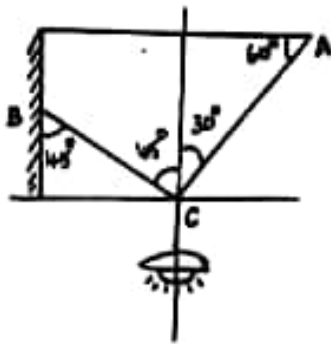
$T_{AB} \sin 50^\circ + T_{AC} \sin 30^\circ - 392.4 = 0$

$T_{AB} \sin 50^\circ + T_{AC} \sin 30^\circ = 392.4 \text{ N} \quad \text{--- (2)}$

On solving (1) & (2), $\Rightarrow T_{AB} = 335.73 \text{ N}$

$T_{AC} = 270.42 \text{ N}$

- (5) (PV) An electrical light fixture weighing 15N hangs from point C, by two strings AC and BC. AC is inclined at 60° to the horizontal and BC at 45° to the vertical as shown in figure. Using Lami's theorem determine the forces in the string AC and BC.



Applying Lami's theorem,

$$\frac{15}{\sin 75} = \frac{T_{BC}}{\sin 150} = \frac{T_{AC}}{\sin 135}$$

On equating, $\frac{T_{BC}}{\sin 150} = \frac{15}{\sin 75}$

$$T_{BC} = \frac{15}{\sin 75} \times \sin 150$$

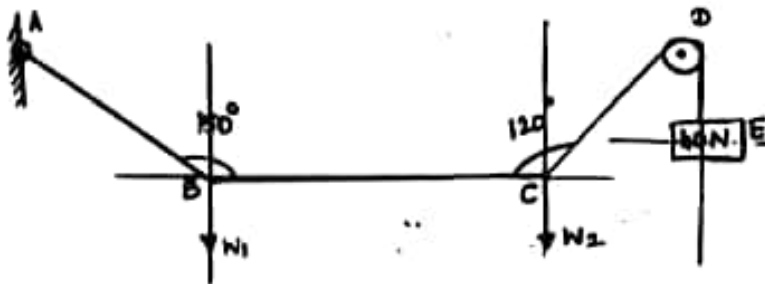
$$T_{BC} = 7.76 \text{ N}$$

On equating, $\frac{T_{AC}}{\sin 135} = \frac{15}{\sin 75}$

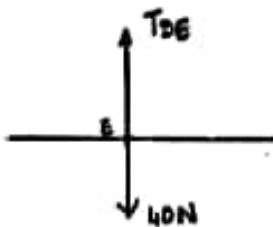
$$T_{AC} = \frac{15}{\sin 75} \times \sin 135$$

$$T_{AC} = 10.98 \text{ N}$$

- Q6 A fine light string ABCDE where extremity A is fixed, has weights w_1 and w_2 attached to it at B and C. It passes over a small smooth peg at D carrying a weight of 40N at the free end E as shown in figure. If in the position of equilibrium, BC is horizontal and AB and CD makes 150° and 120° with BC. Find (a) Tension in the portion AB, BC and CD of the string & (b) Magnitude of w_1 and w_2 .



FBD at E:



$$\sum F_x = 0$$

$$\sum F_y = 0$$

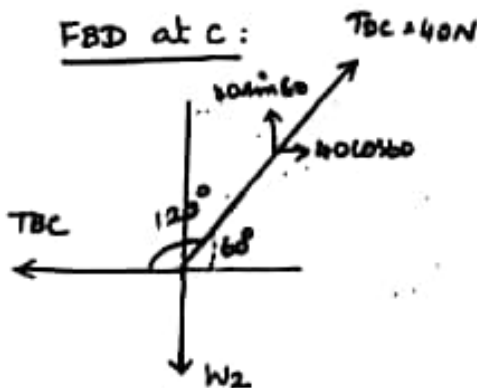
$$T_{DE} - 40 = 0$$

$$T_{DE} = 40\text{N}$$

$$T_{DE} = T_{CD} = 40\text{N}$$

[$\because$ One rope which moves over the pulley has only one tension]

FBD at C:



$$\sum F_x = 0$$

$$40 \cos 60^\circ - T_{BC} = 0$$

$$T_{BC} = 40 \cos 60^\circ$$

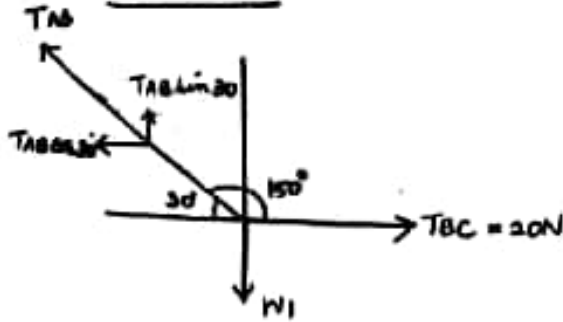
$$T_{BC} = 20\text{N}$$

$$\sum F_y = 0$$

$$40 \sin 60^\circ - w_2 = 0$$

$$w_2 = 34.64\text{N}$$

FBD at B:



$$\sum F_x = 0$$

$$20 - TAB \cos 30^\circ = 0$$

$$TAB \cos 30^\circ = 20$$

$$TAB = \frac{20}{\cos 30^\circ}$$

$$TAB = 23.09 \text{ N}$$

$$\sum F_y = 0$$

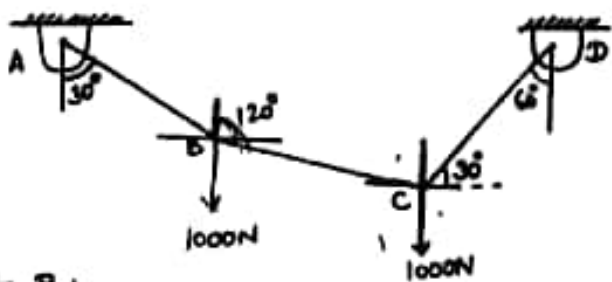
$$TAB \sin 30^\circ - W_1 = 0$$

$$W_1 = TAB \sin 30^\circ$$

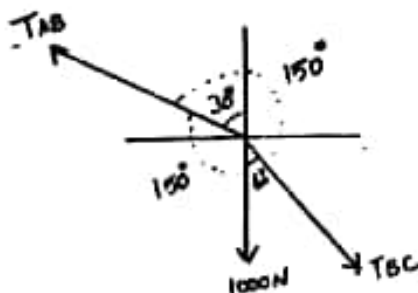
$$= 23.09 \sin 30^\circ$$

$$W_1 = 11.55 \text{ N}$$

- ⑦ A string ABCD attached to two fixed points A and D has two equal weights of 1000N attached to it B and C. The weight rest with the portions AB and CD inclined at angles of 30° and 60° respectively to the vertical as shown in figure. Find the tension in the portions AB, BC and CD of the string if the inclination of the portion BC with vertical is 120°



FBD at B:



Applying Lami's theorem,

$$\frac{1000}{\sin 150^\circ} = \frac{TAB}{\sin 60^\circ} = \frac{TBC}{\sin 150^\circ}$$

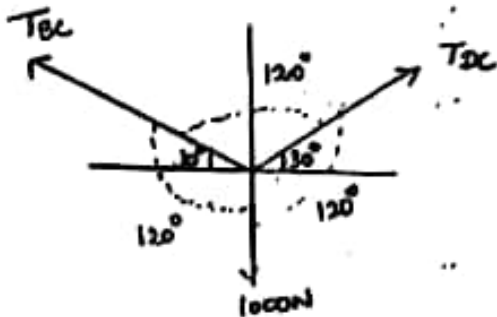
On equating, $\frac{T_{AB}}{\sin 60} = \frac{1000}{\sin 150}$

$T_{AB} = 1732 \text{ N}$

On equating, $\frac{T_{BC}}{\sin 150} = \frac{1000}{\sin 150}$

$T_{BC} = 1000 \text{ N}$

FBD at c:

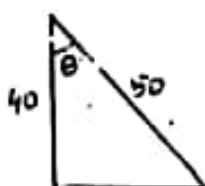
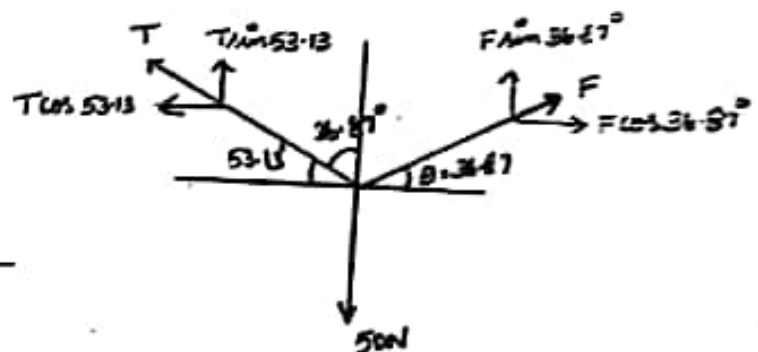
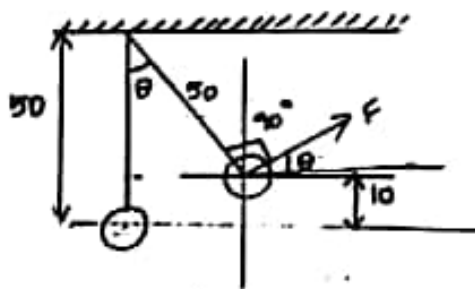


Applying Lami's theorem,

$$\frac{1000}{\sin 120} = \frac{T_{BC}}{\sin 120} = \frac{T_{DC}}{\sin 120}$$

$T_{BC} = T_{DC} = 1000 \text{ N}$

- ⑧ A spherical ball of weight 50N is suspended vertically by a string 50cm long as shown in figure. Find the magnitude and direction of the least force which can hold the ball 10cm above the lowest point. Also find the tension in the string at that point.



$$\cos \theta = \frac{50-10}{50} = \frac{40}{50}$$

$$\theta = \cos^{-1} (40/50)$$

$$\theta = 36.87^\circ$$

Applying resolution of forces,

$$\sum F_x = 0$$

$$F \cos 36.87 - T \cos 53.13 = 0 \quad \text{--- (1)}$$

$$\sum F_y = 0$$

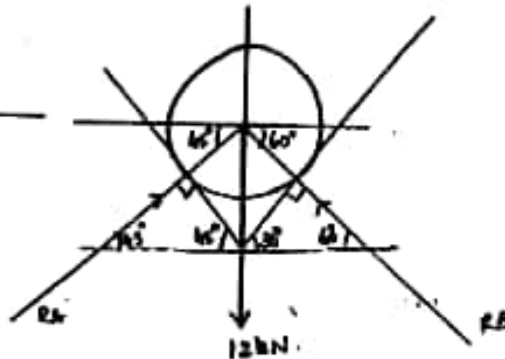
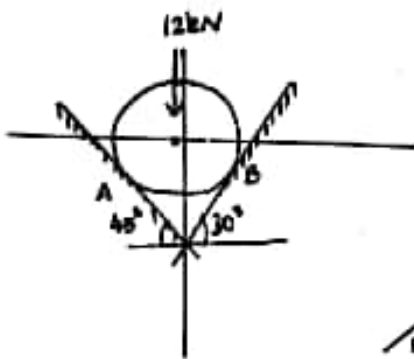
$$F \sin 36.87 + T \sin 53.13 - 50 = 0$$

$$F \sin 36.87 + T \sin 53.13 = 50 \quad \text{--- (2)}$$

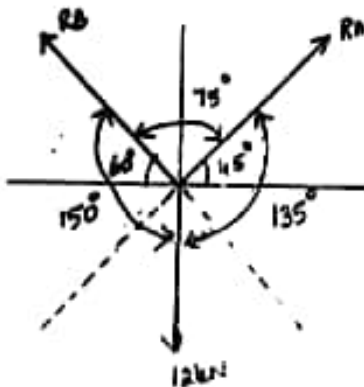
On solving (1) & (2) $\Rightarrow F = 30\text{N}$

$$T = 40\text{N}$$

- ⑨ A heavy spherical ball of weight 12kN rest in a V-shaped container as shown in figure. Find the reaction, assuming the contact surface to be smooth.



FBD:



Applying Lami's theorem,

$$\frac{12}{\sin 75} = \frac{R_A}{\sin 150} = \frac{R_B}{\sin 135}$$

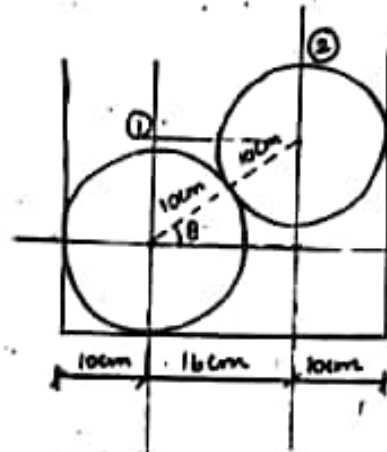
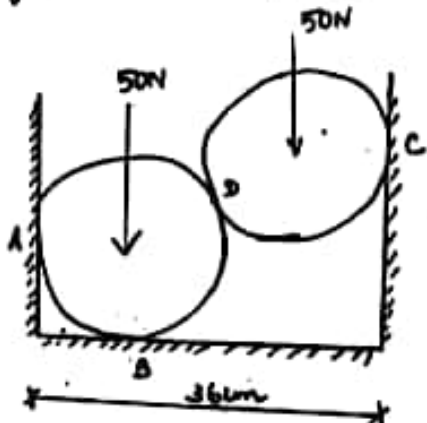
On equating, $\frac{R_A}{\sin 150} = \frac{12}{\sin 75}$

$$R_A = 6.21\text{kN}$$

On equating, $\frac{R_B}{\sin 135} = \frac{12}{\sin 75}$

$R_B = 8.78 \text{ kN}$

- 18) Two identical rollers each of weight 50N and radius 10cm rest in a horizontal channel of width 36cm as shown in figure. Find the reactions at the point of contact A, B, C.

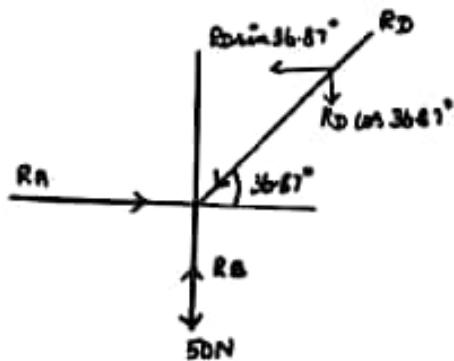


$\cos \theta = \frac{16}{10+10} = \frac{11}{21}$

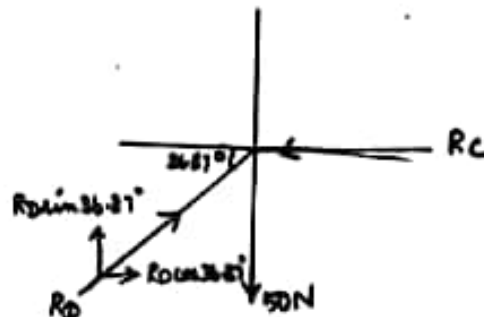
$\theta = \cos^{-1}(16/20)$

$\theta = 36.87^\circ$

FBD for ①:



FBD for ②:



Considering for ①,

$\sum F_x = 0$

$R_A - R_D \cos 36.87 = 0$ — ①

$\sum F_y = 0$

$R_B - R_D \sin 36.87 - 50 = 0$ — ②

Considering for (2),

$$\sum F_x = 0$$

$$-R_C + R_D \cos 36.87^\circ = 0 \quad \text{--- (3)}$$

$$\sum F_y = 0$$

$$R_D \sin 36.87^\circ - 50 = 0$$

$$R_D = \frac{50}{\sin 36.87^\circ}$$

$$R_D = 83.35 \text{ N}$$

$$\text{Sub-| } R_D \text{ in (3)} \Rightarrow -R_C + 83.35 \cos 36.87^\circ = 0$$

$$R_C = 83.35 \cos 36.87^\circ$$

$$R_C = 66.68 \text{ N}$$

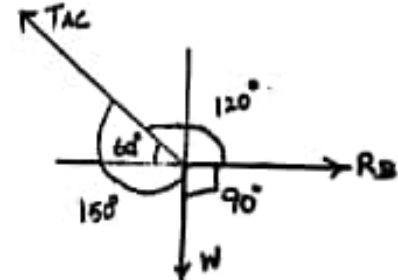
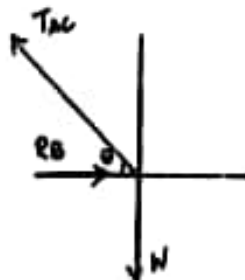
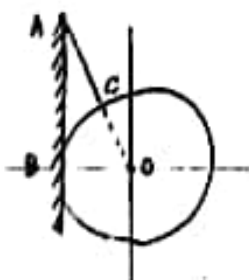
$$\text{Sub-| } R_D \text{ in (1)} \Rightarrow R_A - 83.35 \cos 36.87^\circ$$

$$R_A = 66.68 \text{ N}$$

$$\text{Sub-| } R_D \text{ in (2)} \Rightarrow R_B - 83.35 \cos 36.87^\circ - 50 = 0$$

$$R_B = 99.99 \text{ N}$$

- II A smooth sphere of weight W is supported by a string fastened to a point A on the smooth vertical wall, the other end is in contact with point B on the wall as shown in figure. If the length of the string AC is equal to the radius of the sphere. Find the tension in the string and the reaction of the wall.



Let the radius of sphere $OB = OC = r$

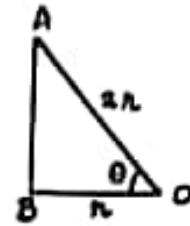
Length of string $AC = \text{radius} = r$

In triangle AOB ,

$$OA = OC + CA = r + r = 2r$$

$$OB = r.$$

$$\cos \theta = \frac{r}{2r} = \frac{1}{2} \Rightarrow \theta = 60^\circ$$



Applying Lami's theorem,

$$\frac{W}{\sin 120} = \frac{T_{AC}}{\sin 90} = \frac{R_B}{\sin 150}$$

On equating,

$$\frac{T_{AC}}{\sin 90} = \frac{W}{\sin 120}$$

$$T_{AC} = \frac{W}{\sin 120} \times \sin 90$$

$$T_{AC} = 1.155W$$

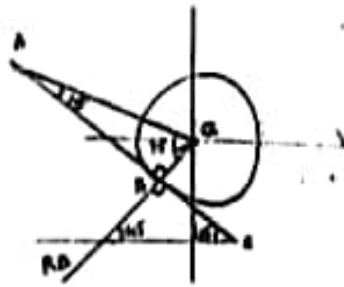
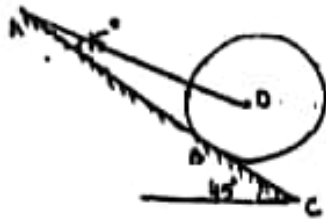
On equating,

$$\frac{R_B}{\sin 150} = \frac{W}{\sin 120}$$

$$R_B = \frac{W}{\sin 120} \times \sin 150$$

$$R_B = 0.577W$$

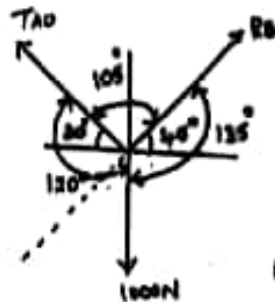
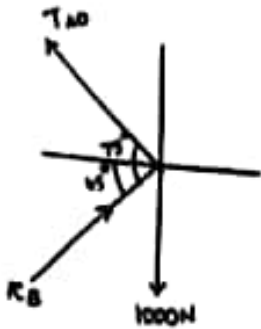
- ⑫ String AO holds a smooth sphere on an inclined plane ABC as shown in figure. The weight of the sphere is $1000N$. Calculate the tension in the string and the reaction at that point of Contact B .



$$\angle QAB + \angle ABO + \angle AOB = 180^\circ$$

$$15^\circ + 90^\circ + \angle AOB = 180^\circ$$

$$\angle AOB = 75^\circ$$



Applying Lami's theorem,

$$\frac{1000}{\sin 105^\circ} = \frac{T_{AO}}{\sin 135^\circ} = \frac{R_B}{\sin 120^\circ}$$

On equating,

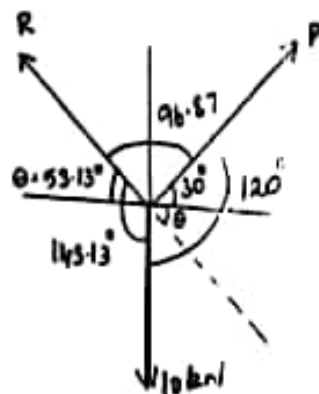
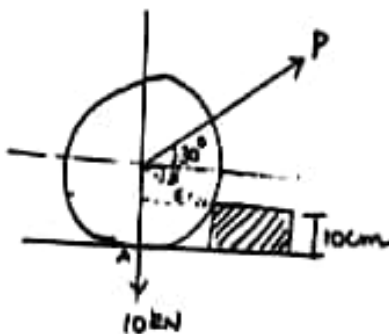
$$\frac{T_{AO}}{\sin 135} = \frac{1000}{\sin 105}$$

$$T_{AO} = 732 \text{ N}$$

On equating, $\frac{R_B}{\sin 120} = \frac{1000}{\sin 105}$

$$R_B = 896.57 \text{ N}$$

- 13) A cylindrical roller has a weight of 10 kN & it is being pulled by a force which is inclined at 30° with the horizontal as shown in figure, while moving it comes across an obstacle 10 cm high. Calculate the force required to cross this obstacle, if the diameter of the roller is 1 m.



$$\sin \theta = \frac{40}{70}$$

$$\theta = 53.13^\circ$$

NOTE: R_A should not be considered, because when roller crosses over the obstacle, R_A is absent

Applying Lami's theorem,

$$\frac{10}{\sin 96.87} = \frac{R}{\sin 120} = \frac{P}{\sin 143.13}$$

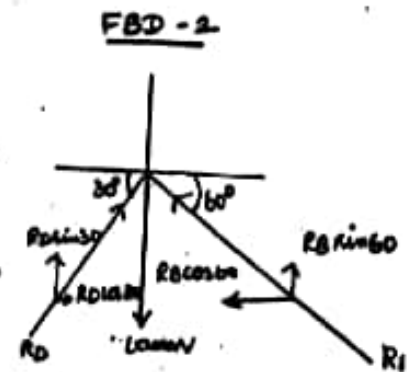
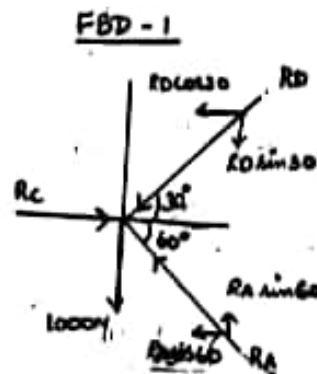
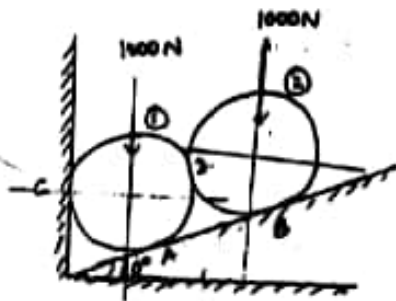
On equating, $\rightarrow \frac{R}{\sin 120} = \frac{10}{\sin 96.87}$

$$R = 8.72 \text{ kN}$$

On equating $\rightarrow \frac{P}{\sin 143.13} = \frac{10}{\sin 96.87}$

$$P = 6.04 \text{ kN}$$

- 14) Two identical rollers each of weight $W = 1000 \text{ N}$ are supported by an inclined plane and a vertical wall as shown in figure. Find the reactions at the points of supports A, B & C.



Considering FBD - ①,

$$\sum F_x = 0$$

$$R_c - R_D \cos 30^\circ - R_A \cos 60^\circ = 0 \quad \text{--- (1)}$$

$$\sum F_y = 0$$

$$-R_D \sin 30^\circ - 1000 + R_A \sin 60^\circ = 0 \quad \text{--- (2)}$$

Considering FBD (2),

$$\sum F_x = 0$$

$$R_D \cos 30^\circ - R_B \cos 60^\circ = 0 \quad \text{--- (3)}$$

$$\sum F_y = 0$$

$$R_D \sin 30^\circ + R_B \sin 60^\circ - 1000 = 0$$

$$R_D \sin 30^\circ + R_B \sin 60^\circ = 1000 \quad \text{--- (4)}$$

Solving (3) & (4) $\Rightarrow R_D = 500N$

$$R_B = 866.03N$$

Sub-| R_D in (1) $\Rightarrow -500 \sin 30^\circ - 1000 + R_A \sin 60^\circ = 0$

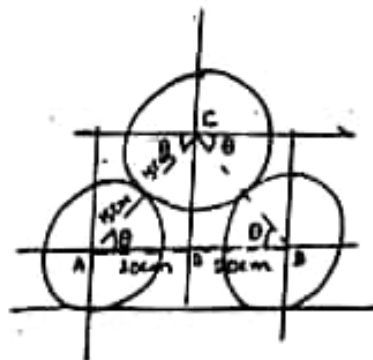
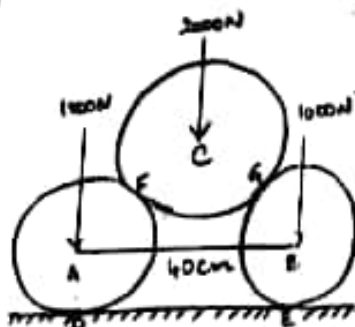
$$R_A = \frac{1250}{\sin 60^\circ}$$

$$R_A = 1443.38N$$

(1) $\Rightarrow R_C - 500 \cos 30^\circ - 1443.38 \cos 60^\circ = 0$

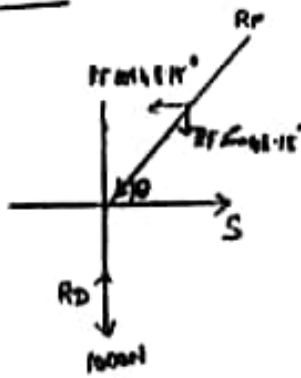
$$R_C = 1154.70N$$

- (14) Two smooth cylinders, each of weight $W = 1000N$ and radius $15cm$ are connected at their centers by a string AB of length $= 40cm$ and rest upon a horizontal plane, supporting above them a third cylinder of weight $= 2000N$ and radius $15cm$ as shown in figure. Find the force S in the string AB and the pressure produced on the floor.

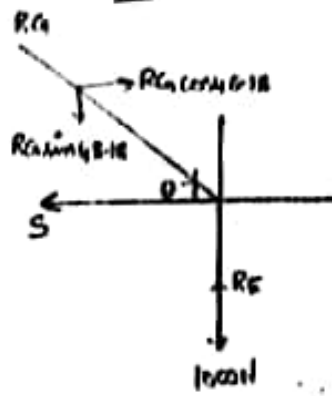


$\cos \theta = \frac{20}{30}$
 $\theta = \cos^{-1} \left(\frac{2}{3} \right)$
 $\theta = 48.18^\circ$

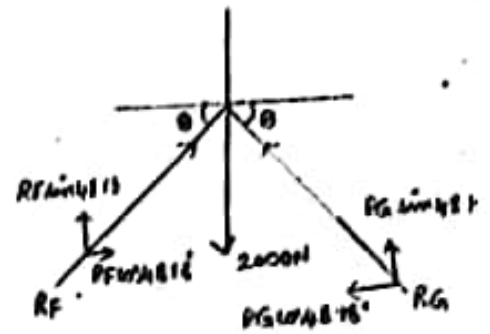
FBD - A



FBD - B



FBD - C



Considering FBD-A, $\sum F_x = 0$

$$S - R_F \cos 48.18^\circ = 0 \quad \text{--- (1)}$$

$$\sum F_y = 0$$

$$R_D - 1000 - R_F \sin 48.18^\circ = 0 \quad \text{--- (2)}$$

Considering FBD-B, $\sum F_x = 0$

$$R_G \cos 48.18^\circ - S = 0 \quad \text{--- (3)}$$

$$\sum F_y = 0$$

$$-R_G \sin 48.18^\circ + R_E - 1000 = 0 \quad \text{--- (4)}$$

Considering FBD-C, $\sum F_x = 0$

$$R_F \cos 48.18^\circ - R_G \cos 48.18^\circ = 0 \quad \text{--- (5)}$$

$$\sum F_y = 0$$

$$R_F \sin 48.18^\circ + R_G \sin 48.18^\circ - 2000 = 0$$

$$R_F \sin 48.18^\circ + R_G \sin 48.18^\circ = 2000 \quad \text{--- (6)}$$

$$\text{Solving (5) \& (6)} \Rightarrow R_F = 1341.84 \text{ N}$$

$$R_G = 1341.84 \text{ N}$$

$$\text{Sub in eqn (4)} \Rightarrow -1341.84 \sin 48.18^\circ + R_E - 1000 = 0$$

$$R_E = 1000 + 1341.84 \sin 48.18^\circ$$

$$R_E = 1999.99 \approx R_E = 2000 \text{ N}$$

Sub-| R_G in (3) $\Rightarrow 1341.84 \cos 48.18 - S = 0$

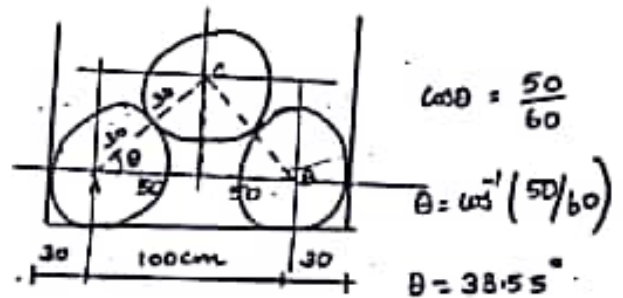
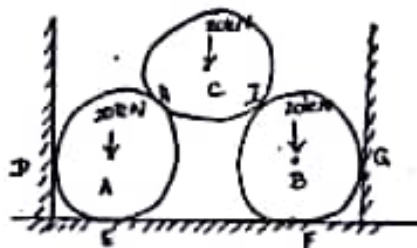
$S = 894.73 \text{ N}$

Sub-| R_F in (2) $\Rightarrow R_D - 1000 - 1341.84 \sin 48.18 = 0$

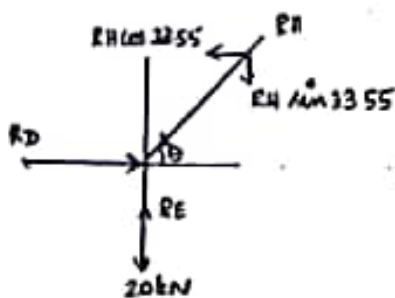
$R_D = 1000 + 1341.84 \sin 48.18$

$R_D = 1999.99 \approx R_D = 2000 \text{ N}$

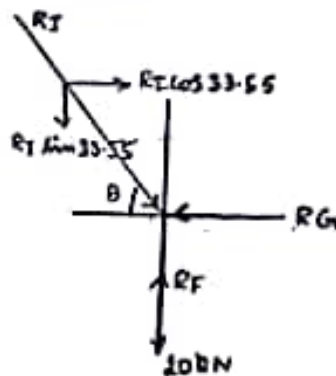
- (15) Three smooth pipes each weighing 20 kN and of diameter 60 cm are to be placed in a rectangular channel with horizontal base as shown in figure. Calculate the reactions at the point of contact. Take width of channel as 160 cm .



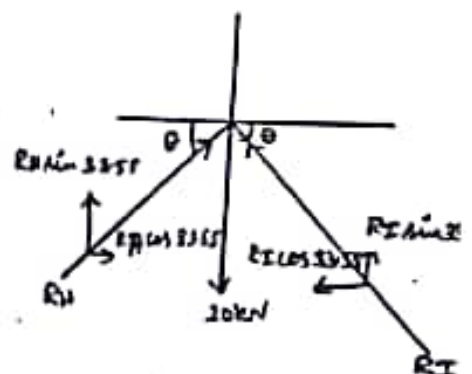
FBD - A



FBD - B



FBD - C



Considering FBD - A,

$\sum F_x = 0$

$R_D - R_H \cos 33.55 = 0 \quad \text{--- (1)}$

$\sum F_y = 0$

$-R_H \sin 33.55 + R_E - 20 = 0 \quad \text{--- (2)}$

Considering FBD - B, $\sum F_x = 0$

$$R_I \cos 33.55 - R_G = 0 \quad \text{--- (3)}$$

$$\sum F_y = 0$$

$$- R_I \cos 33.55 - 20 + R_F = 0 \quad \text{--- (4)}$$

Considering FBD - C,

$$\sum F_x = 0$$

$$R_H \cos 33.55 - R_I \cos 33.55 = 0 \quad \text{--- (5)}$$

$$\sum F_y = 0$$

$$R_H \sin 33.55 + R_I \sin 33.55 - 20 = 0$$

$$R_H \sin 33.55 + R_I \sin 33.55 = 20 \quad \text{--- (6)}$$

$$\text{Solving (5) \& (6)} \Rightarrow \begin{aligned} R_H &= 18.09 \text{ kN} \\ R_I &= 18.09 \text{ kN} \end{aligned}$$

$$\text{Sub } R_H \text{ in (1)} \Rightarrow R_D - 18.09 \cos 33.55 = 0$$

$$R_D = 15.07 \text{ kN}$$

$$\text{Sub } R_H \text{ in (2)} \Rightarrow -18.09 \sin 33.55 + R_E - 20 = 0$$

$$R_E = 30 \text{ kN}$$

$$\text{Sub } R_I \text{ in (3)} \Rightarrow 18.09 \cos 33.55 - R_G = 0$$

$$R_G = 15.07 \text{ kN}$$

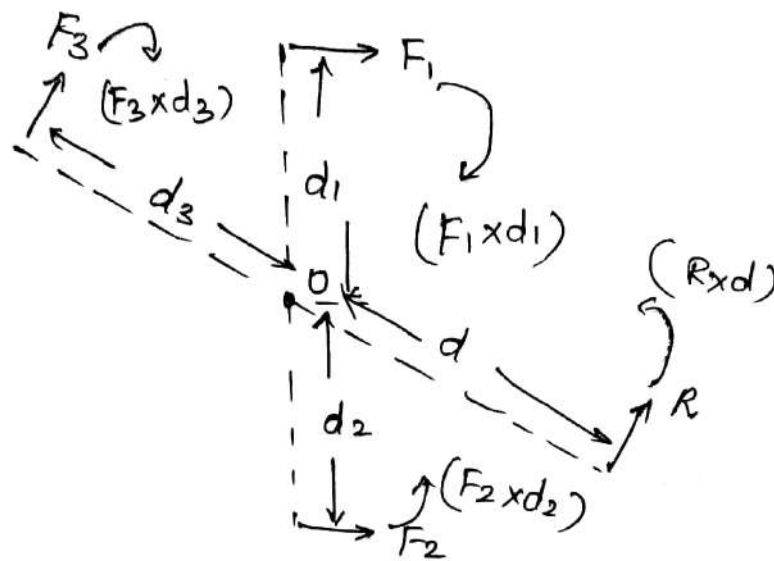
$$\text{Sub } R_I \text{ in (4)} \Rightarrow -18.09 \cos 33.55 - 20 + R_F = 0$$

$$R_F = 30 \text{ kN}$$

VARIGNON'S THEOREM

Statement:

The algebraic sum of moment of all force about a point 'o' which is equal to the moment of resultant force about the same point



$$\sum M_o = R \times d$$

$$(F_1 \times d_1) + (F_2 \times d_2) + (F_3 \times d_3) = R \times d$$

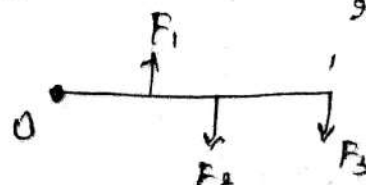
$$d = ?$$

Where d is the location of resultant force.

Application:

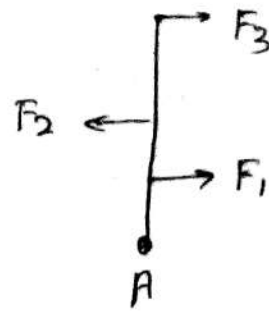
1. To locate the resultant force
2. If the force is acting on a horizontal rod

$$\sum M_o = R \times x$$



3. If the forces are acting in a vertical rod

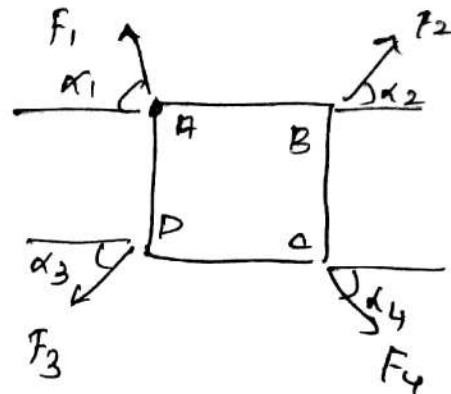
$$\sum M_A = R \times y$$



4. If the forces are acting on the vertical and horizontal plane
A-reference point

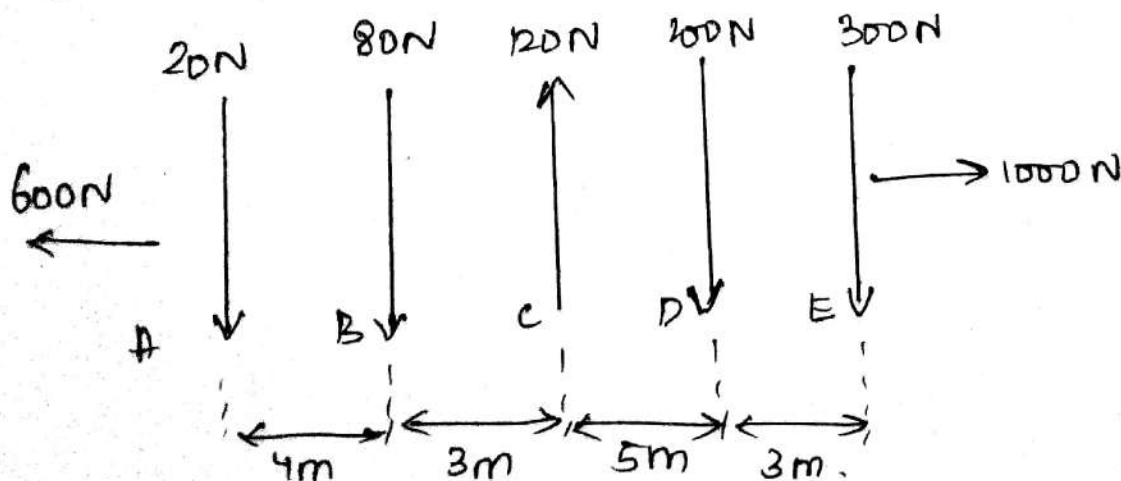
$$\sum M_A = \sum F_V \times x$$

$$\sum M_A = \sum F_H \times y$$



Problems:

1. Find the resultant of the force and its position from A for the given system as shown in fig



Soln:

$$\begin{aligned}\sum F_V &= -20 - 80 + 120 - 200 - 300 \\ &= -480 \text{ N}\end{aligned}$$

$$\begin{aligned}\sum F_H &= -600 + 1000 \text{ N} \\ &= 400 \text{ N}\end{aligned}$$

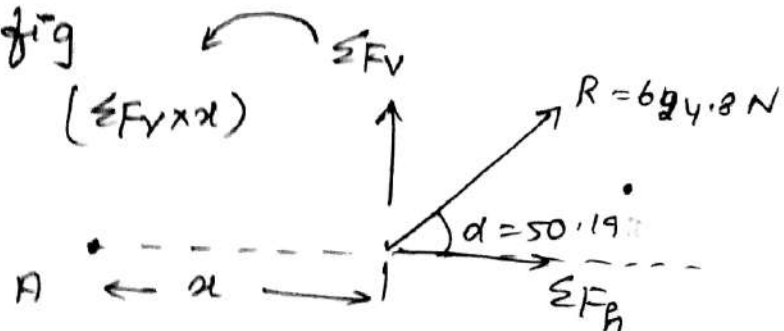
$$\begin{aligned}R &= \sqrt{\sum F_V^2 + \sum F_H^2} \\ &= \sqrt{(-480)^2 + 400^2}\end{aligned}$$

$$\boxed{R = 624.8 \text{ N}}$$

$$\begin{aligned}\alpha &= \tan^{-1} \left| \frac{\sum F_V}{\sum F_H} \right| \\ &= \tan^{-1} \left| \frac{-480}{400} \right|\end{aligned}$$

$$\boxed{\alpha = 50.19^\circ}$$

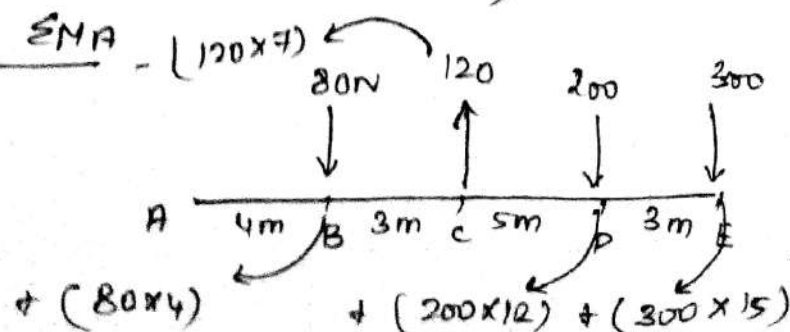
Resultant fig



Varignon's theorem:

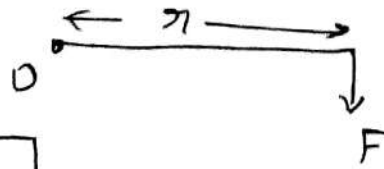
$$\sum M_A = -(\sum F_V \times x)$$

To find $\sum M_A$



Moment:

It is the turning effect of an object produced by the force F about a point 'O'.



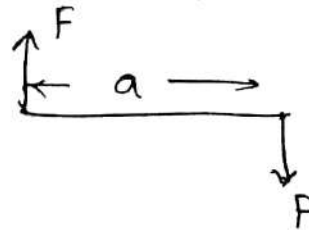
$$M_o = F \times x$$

F = force applied at an object (N)

x = $\perp$ distance b/w O and F
unit for M_o : N-m.

Couple:

Two parallel force having same magnitude but acting in opposite direction.



Moment of the couple = Force $\times$ arm of the couple

$$M_c = F \times a$$

Classification of moment:

1. Clockwise moment



(Positive moment)
+ve

2. Anticlockwise moment



Anticlockwise moment $\rightarrow$ -ve

$$\sum M_A = (80 \times 4) + (200 \times 12) + (300 \times 15) - (120 \times 7)$$

$$\boxed{\sum M_A = 6380 \text{ N-m}}$$

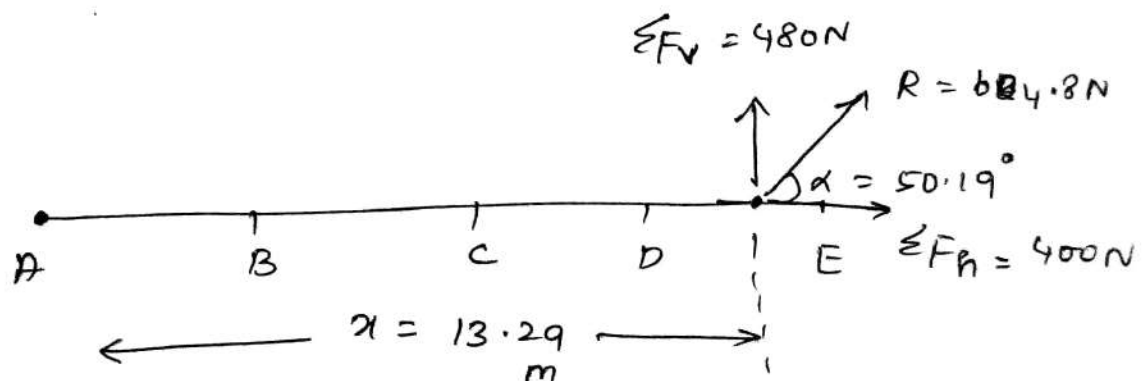
$$\sum M_A = -\sum F_V \times x$$

$$x = \left| \frac{\sum M_A}{\sum F_V} \right|$$

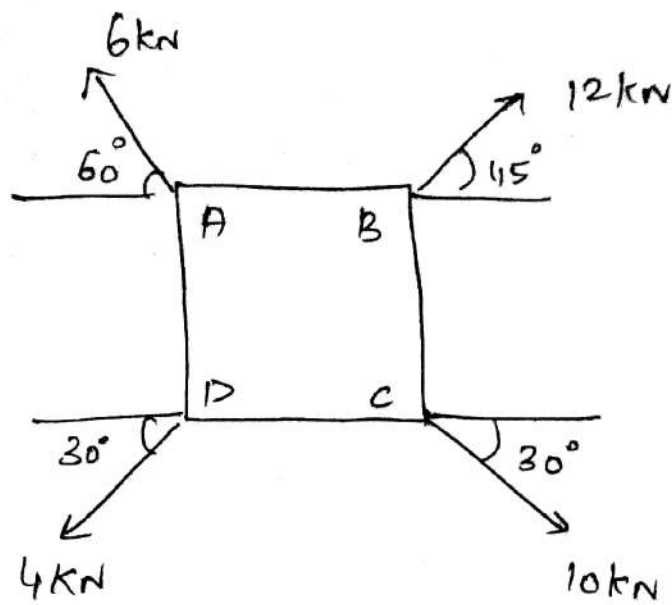
$$= \left| \frac{6380}{480} \right|$$

$$\boxed{x = 13.29 \text{ m}}$$

Ans. fig

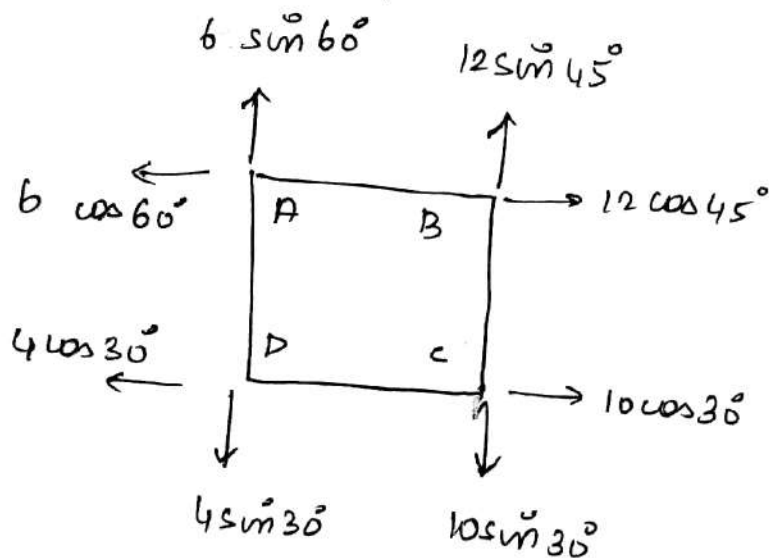


2. Four forces of magnitude and direction acting on a square ABCD of side 2m as shown in fig. Calculate the resultant in magnitude and direction and also locate the point of application with respect to the sides AB and AD.



Solution:

1. Resolution fig:



$$\sum F_y = 12 \sin 45^\circ + 6 \sin 60^\circ - 4 \sin 30^\circ - 10 \sin 30^\circ$$

$$= 6.68 \text{ kN}$$

$$\sum F_x = 12 \cos 45^\circ - 6 \cos 60^\circ - 4 \cos 30^\circ + 10 \cos 30^\circ$$

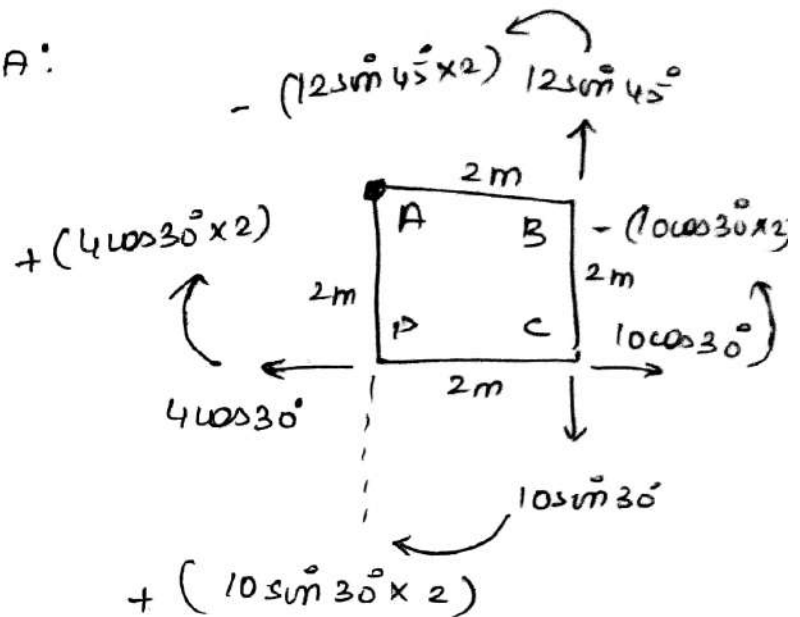
$$= 10.68 \text{ kN}$$

$$R = \sqrt{\sum F_y^2 + \sum F_x^2} = 12.5 \text{ kN}$$

$$\alpha = \tan^{-1} \left| \frac{\sum F_V}{\sum F_H} \right|$$

$$\boxed{\alpha = 32.02^\circ}$$

2. To find $\sum M_A$:



$$\sum M_A = -(12.5 \sin 45^\circ \times 2) - (10 \cos 30^\circ \times 2) + (10 \sin 30^\circ \times 2) + (4 \cos 30^\circ \times 2)$$

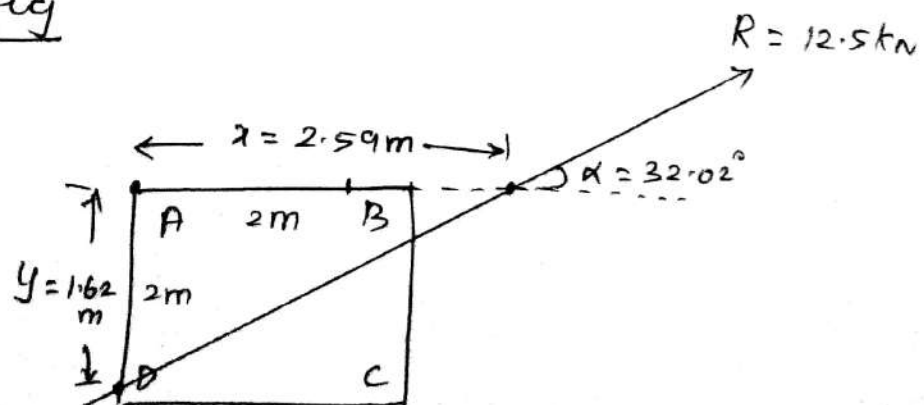
$$\boxed{\sum M_A = -17.36 \text{ kN-m}}$$

3. To find x & y

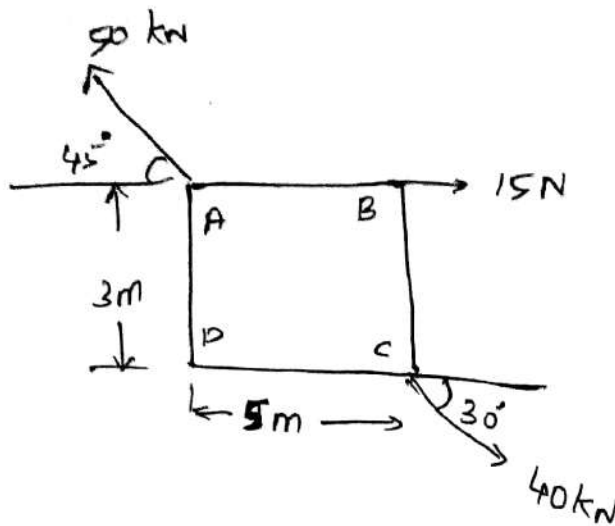
$$x = \left| \frac{\sum M_A}{\sum F_V} \right| = \left| \frac{17.36}{6.68} \right| = 2.59 \text{ m}$$

$$y = \left| \frac{\sum M_A}{\sum F_H} \right| = \left| \frac{17.36}{10.68} \right| = 1.62 \text{ m}$$

4. Ans. fig

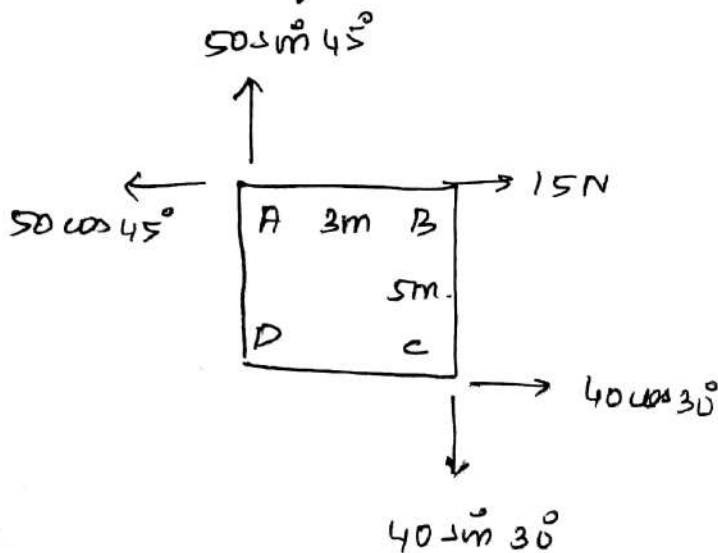


3. Using Varignon's theorem, Determine the magnitude and line of action of the resultant force as shown in fig.



Soln:

1. Resultant force



$$\sum F_H = -50 \cos 45^\circ + 40 \cos 30^\circ + 15$$

$$= 14.28 \text{ kN}$$

$$\sum F_V = 50 \sin 45^\circ - 40 \sin 30^\circ$$

$$= 15.35 \text{ kN}$$

$$R = \sqrt{\sum F_v^2 + \sum F_h^2}$$

$$= \sqrt{15.35^2 + 14.28^2}$$

$$R = 20.96 \text{ kN}$$

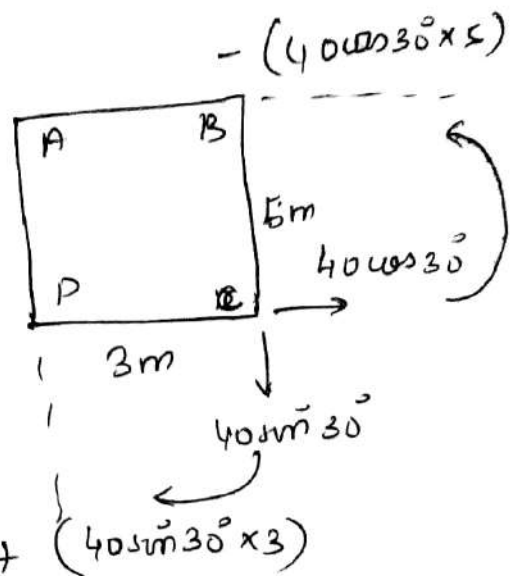
$$\alpha = \tan^{-1} \left| \frac{\sum F_v}{\sum F_h} \right|$$

$$\alpha = 47.06^\circ$$

2. To find $\sum M_A$:

$$\sum M_A = -(40 \cos 30^\circ \times 5) + (40 \sin 30^\circ \times 3)$$

$$\sum M_A = -43.92 \text{ kN}$$

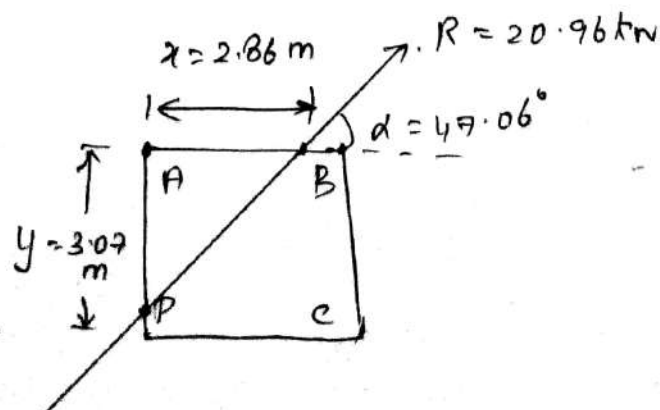


3. To find x and y

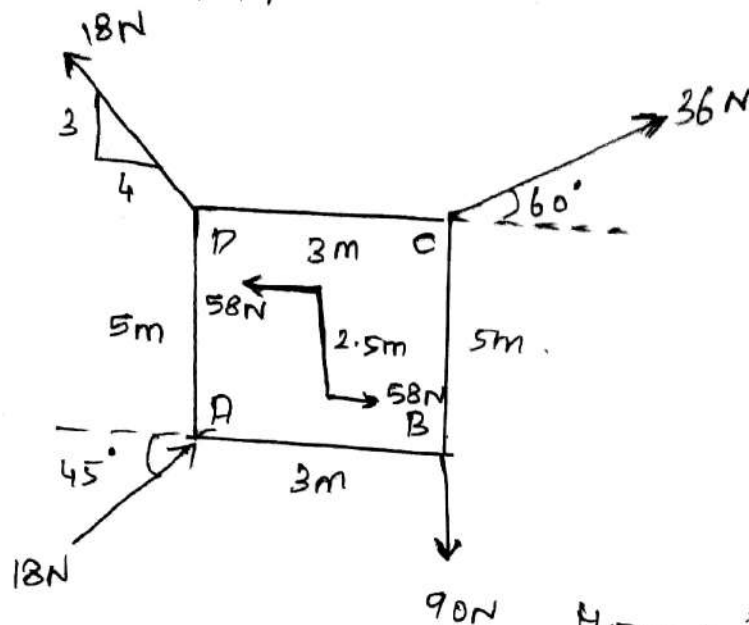
$$x = \left| \frac{\sum M_A}{\sum F_v} \right| = \left| \frac{43.92}{15.35} \right| = 2.86 \text{ m}$$

$$y = \left| \frac{\sum M_A}{\sum F_h} \right| = \left| \frac{43.92}{14.28} \right| = 3.07 \text{ m}$$

4. Ans. fig



4. Determine the resultant force in magnitude and direction and also locate the resultant force with respect to AB and AD.

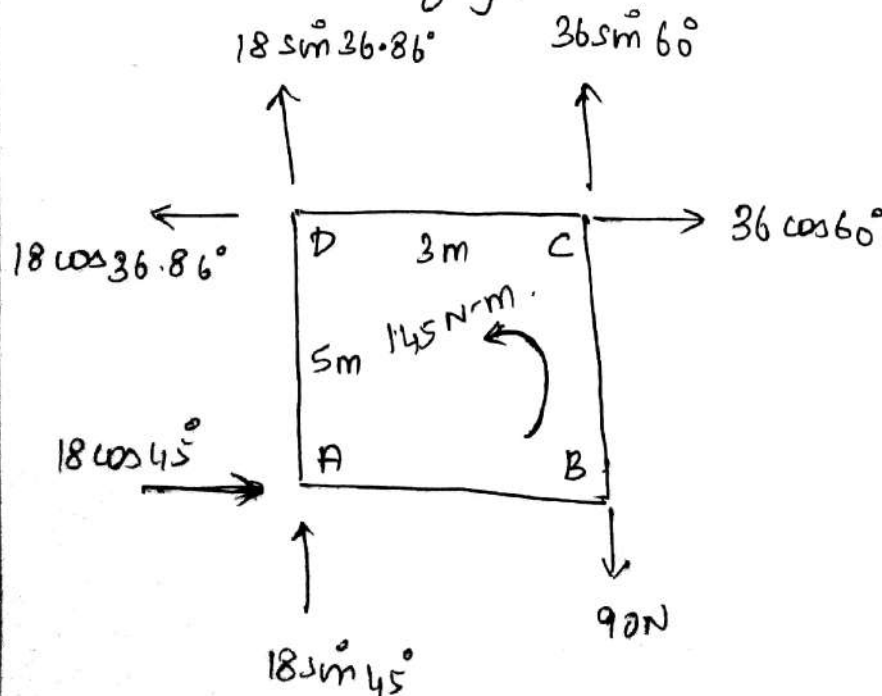


Soln:

$$\tan \theta = 3/4 = 36.86^\circ$$

$$\begin{aligned} \text{Moment} &= F \times a \\ \text{of couple} &= 58 \times 2.5 \\ &= 145 \text{ N-m} \end{aligned}$$

1. Resolution fig:



$$\sum F_y = 18 \sin 36.86^\circ + 36 \sin 60^\circ + 18 \sin 45^\circ - 90$$

$$\sum F_y = -35.29 \text{ N}$$

$$\sum F_H = 36 \cos 60^\circ + 18 \cos 45^\circ - 18 \cos 36.86^\circ$$

$$= 16.32 \text{ N}$$

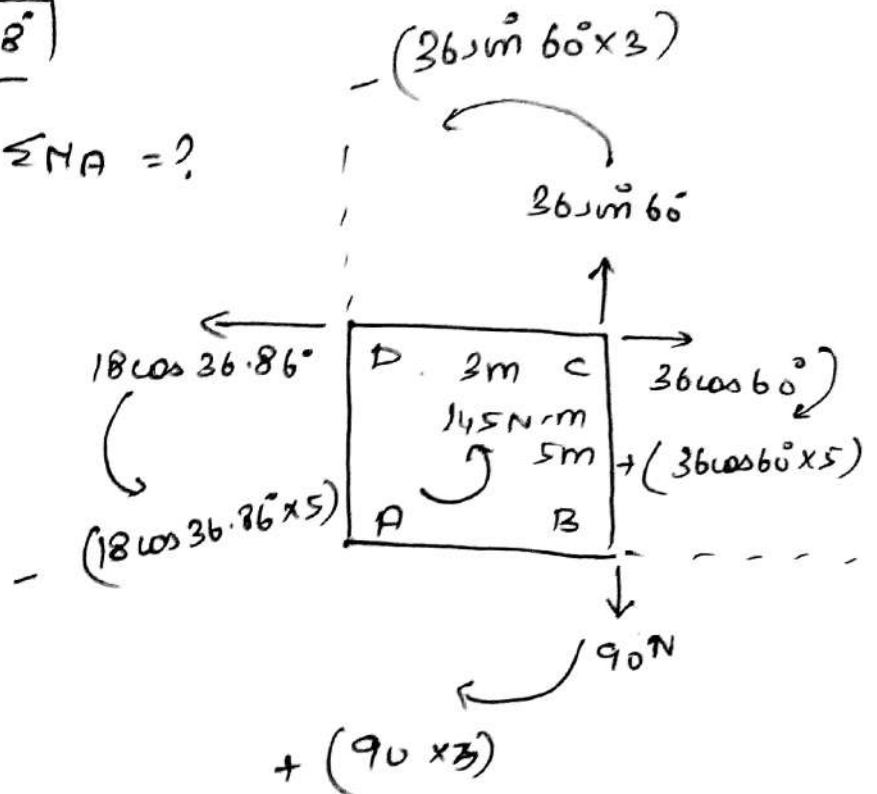
$$R = \sqrt{\sum F_V^2 + \sum F_H^2}$$

$$R = 38.88 \text{ N}$$

$$\alpha = \tan^{-1} \left| \frac{\sum F_V}{\sum F_H} \right|$$

$$\alpha = 65.18^\circ$$

2. To find $\sum M_A = ?$



$$\sum M_A = (90 \times 3) + (36 \cos 60^\circ \times 5) - (36 \sin 60^\circ \times 3)$$

$$- (18 \cos 36.86^\circ \times 5) - 145$$

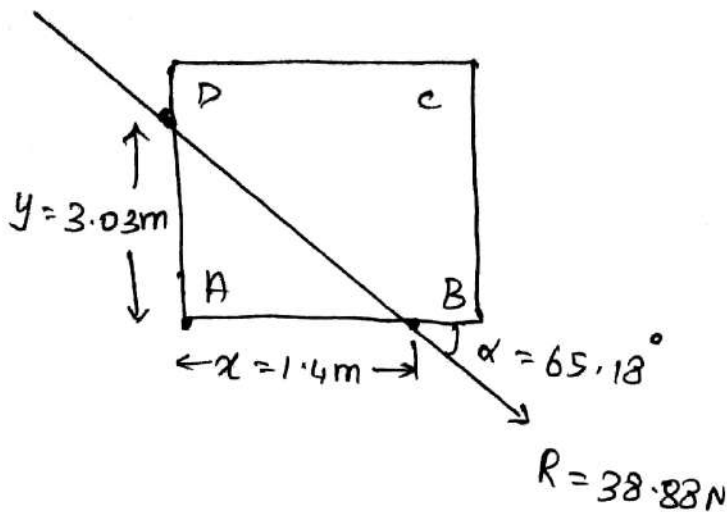
$$\sum M_A = 49.45 \text{ N-m}$$

3. To find x and y

$$x = \left| \frac{\sum M_A}{\sum F_V} \right| = \left| \frac{49.45}{35.29} \right| = 1.40 \text{ m}$$

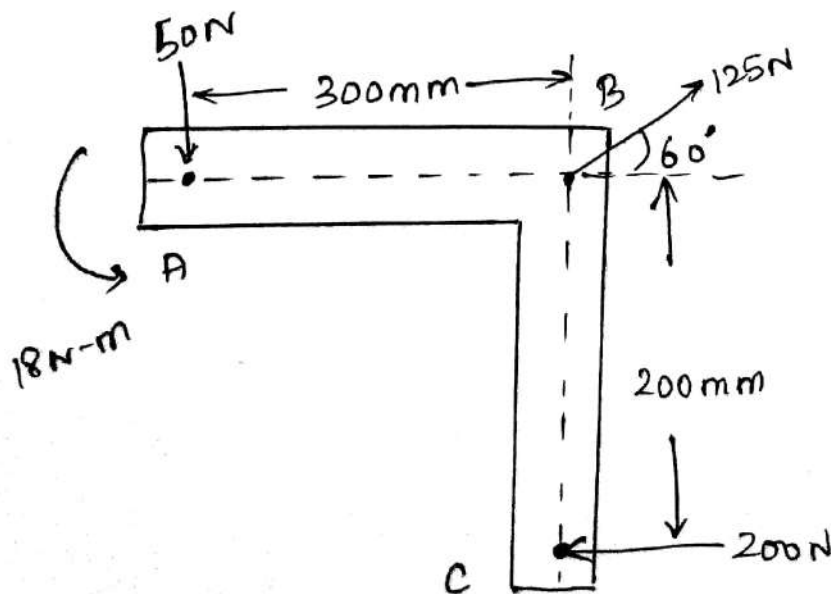
$$y = \left| \frac{\sum M_A}{\sum F_H} \right| = \left| \frac{49.45}{16.32} \right| = 3.03 \text{ m}$$

4. Ans. fig



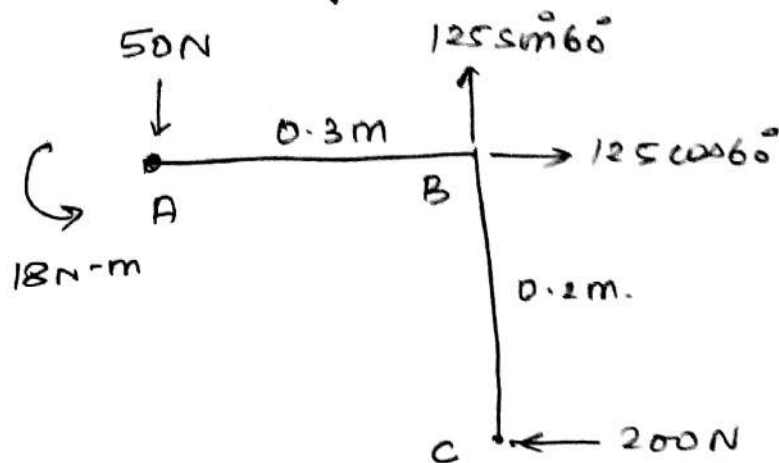
5. Three forces & couple of magnitude $M = 18\text{N-m}$ are applied at an angle section as shown in fig. Calculate

1. Resultant force in magnitude and direction
2. Locate the points where line of action intersects the line AB and BC.



Solution:

1. Resolution fig:



$$\sum F_V = 125 \sin 60^\circ - 50 = 58.25 \text{ N}$$

$$\sum F_H = 125 \cos 60^\circ - 200 = -137.5 \text{ N}$$

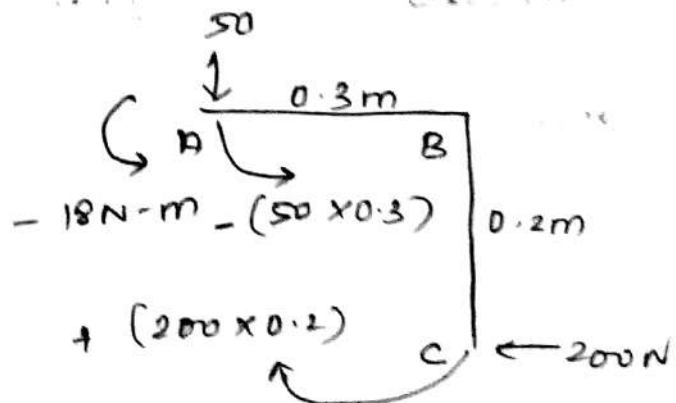
$$R = \sqrt{\sum F_V^2 + \sum F_H^2} = 149.32 \text{ N}$$

$$\alpha = \tan^{-1} \left| \frac{\sum F_V}{\sum F_H} \right|$$

$$\alpha = 22.95^\circ$$

2. To find $\sum M_A$:

$$\sum M_A =$$



$$\sum M_A = (200 \times 0.2) - 18 - (50 \times 0.3)$$

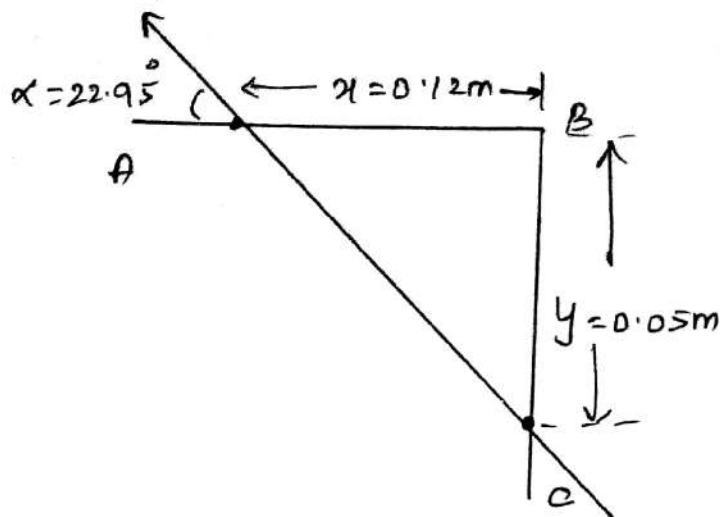
$$\boxed{\sum M_A = 7 \text{ N-m}}$$

3. To find x and y

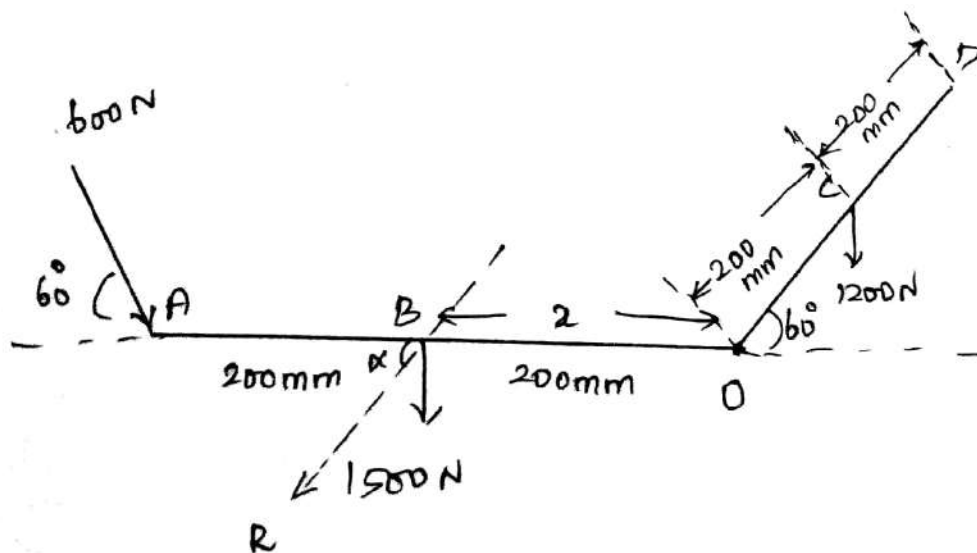
$$x = \left| \frac{\sum M_A}{\sum F_V} \right| = \left| \frac{7}{58.25} \right| = 0.12 \text{ m}$$

$$y = \left| \frac{\sum M_A}{\sum F_H} \right| = \left| \frac{7}{137.5} \right| = 0.05 \text{ m}$$

4. Ans. fig $R = 149.32 \text{ N}$



6. A system of forces acting on a bell crank as shown in fig. Determine the magnitude and direction of point of application of Resultant force about 'O' point using Varignon's theorem.



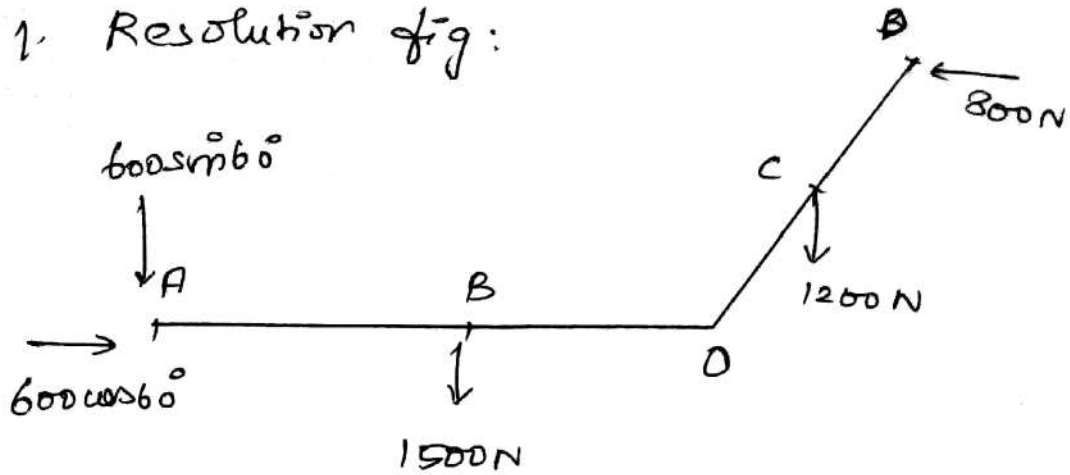
To find: $R = ?$

$\alpha = ?$

$x = ?$

Soln:

1. Resolution fig:



$$\sum F_V = -600 \sin 60^\circ - 1500 - 1200$$

$$= -3219.61 \text{ N}$$

$$\sum F_H = 600 \cos 60^\circ - 800 = -500 \text{ N}$$

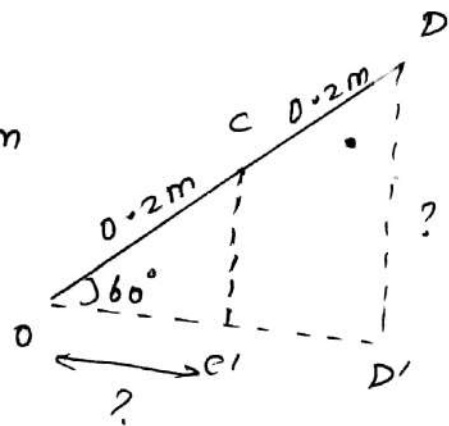
$$R = \sqrt{\sum F_V^2 + \sum F_H^2} = 3258.20 \text{ N}$$

$$\alpha = \tan^{-1} \left| \frac{\sum F_V}{\sum F_H} \right| = 81.17^\circ$$

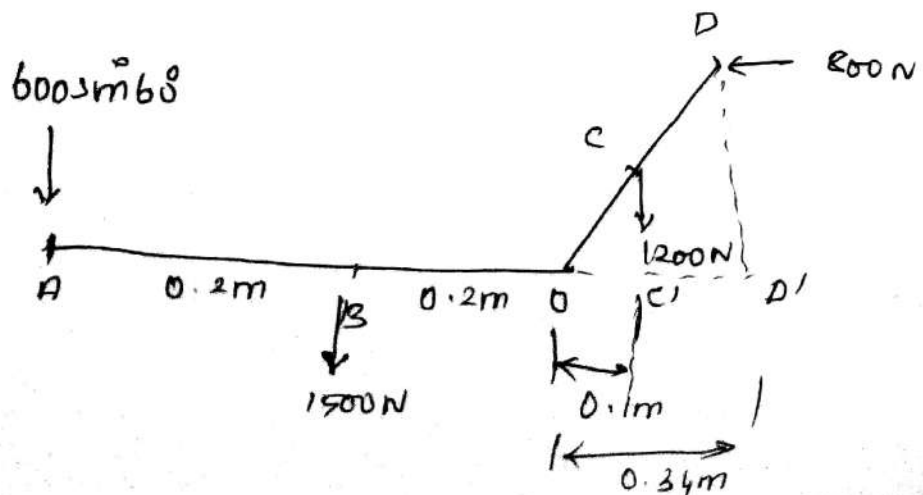
2. To find $\sum M_A$:

$$\cos 60^\circ = \frac{OC'}{OC} \Rightarrow OC' = 0.1 \text{ m}$$

$$\sin 60^\circ = \frac{DD'}{OD} \Rightarrow DD' = 0.34 \text{ m}$$



3.



$$\sum M_A = -(600 \sin 60^\circ \times 0.4) - (1500 \times 0.2) \\ + (1200 \times 0.1) - (800 \times 0.34)$$

$$\sum M_A = -659.84 \text{ N-m}$$

3. To find x & y

$$x = \left| \frac{\sum M_A}{\sum F_V} \right| = \left| \frac{659.84}{3299.61} \right|$$

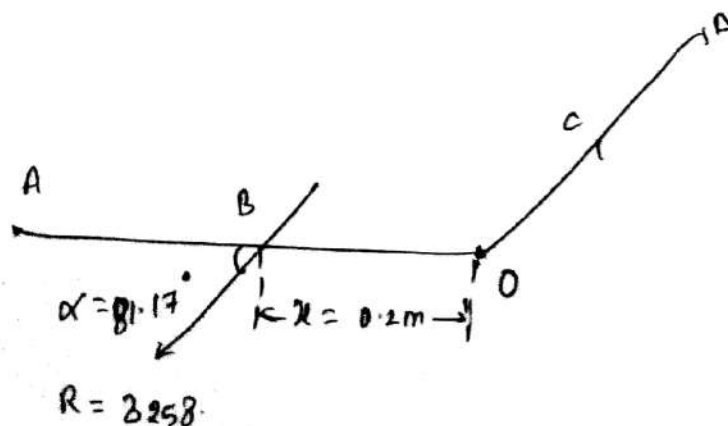
$$x = 0.2 \text{ m}$$

$$y = \left| \frac{\sum M_A}{\sum F_H} \right|$$

$$= \left| \frac{659.84}{500} \right|$$

$$y = 1.31 \text{ m}$$

4. As fig.



$$\sum F_y = 3 \sin 30^\circ - 6 \sin 45^\circ - 5$$

$$= -7.742 \text{ kN}$$

$$\sum F_h = 3 \cos 30^\circ + 6 \cos 45^\circ + 10 + 3 - 8$$

$$= 11.84 \text{ kN}$$

$$R = \sqrt{\sum F_v^2 + \sum F_h^2}$$

$$R = 14.14 \text{ kN}$$

$$\alpha = \tan^{-1} \left| \frac{\sum F_v}{\sum F_h} \right|$$

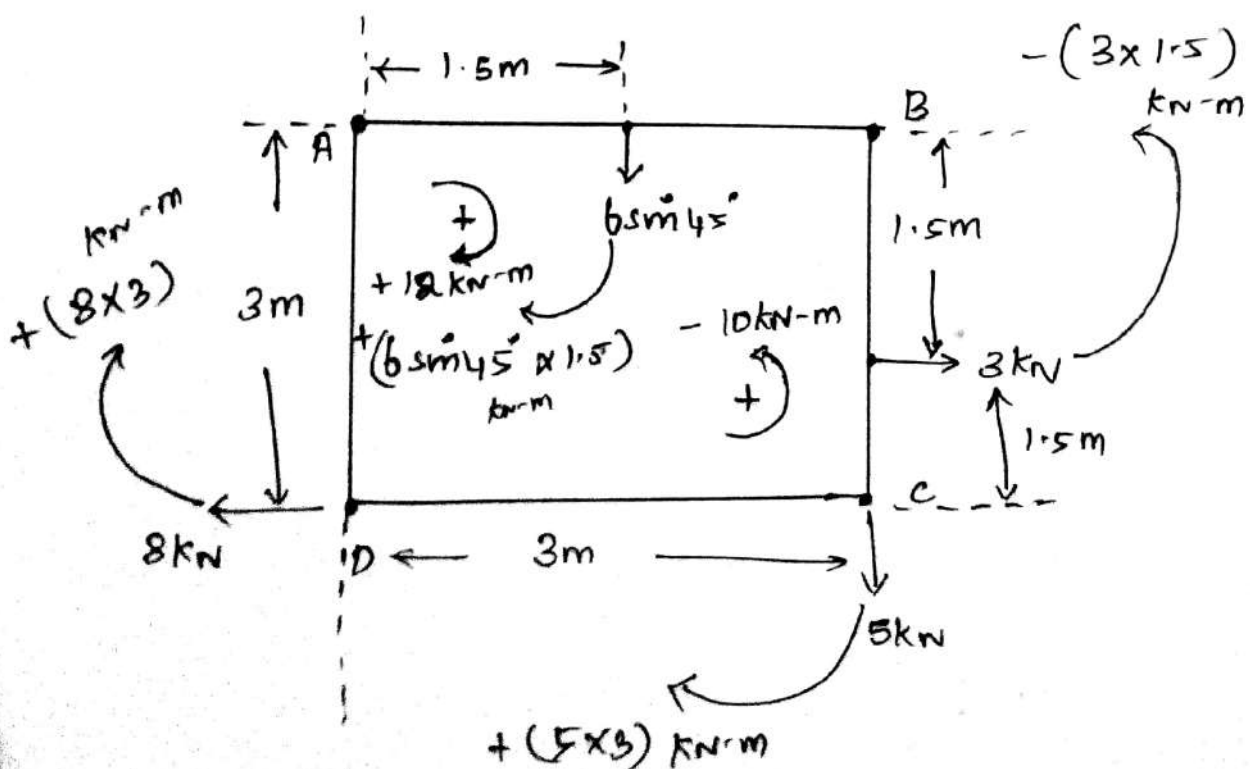
$$\alpha = \tan^{-1} \left| \frac{7.742}{11.84} \right|$$

$$\alpha = 33.18^\circ$$

Step: 2 Finding $\sum MA$.

$$\sum MA = \sum F_v \times x$$

$$\sum MA = \sum F_h \times y$$



$$\Sigma M_A = (8 \times 3) + (5 \times 3) - (3 \times 1.5) + (6 \sin 45^\circ \times 1.5) + 12 - 10$$

$$\Sigma M_A = 42.86 \text{ kN-m [clockwise]}$$

Step: 3 finding x and y

$$x = \left| \frac{\Sigma M_A}{\Sigma F_v} \right|$$

$$= \left| \frac{42.86}{7.742} \right|$$

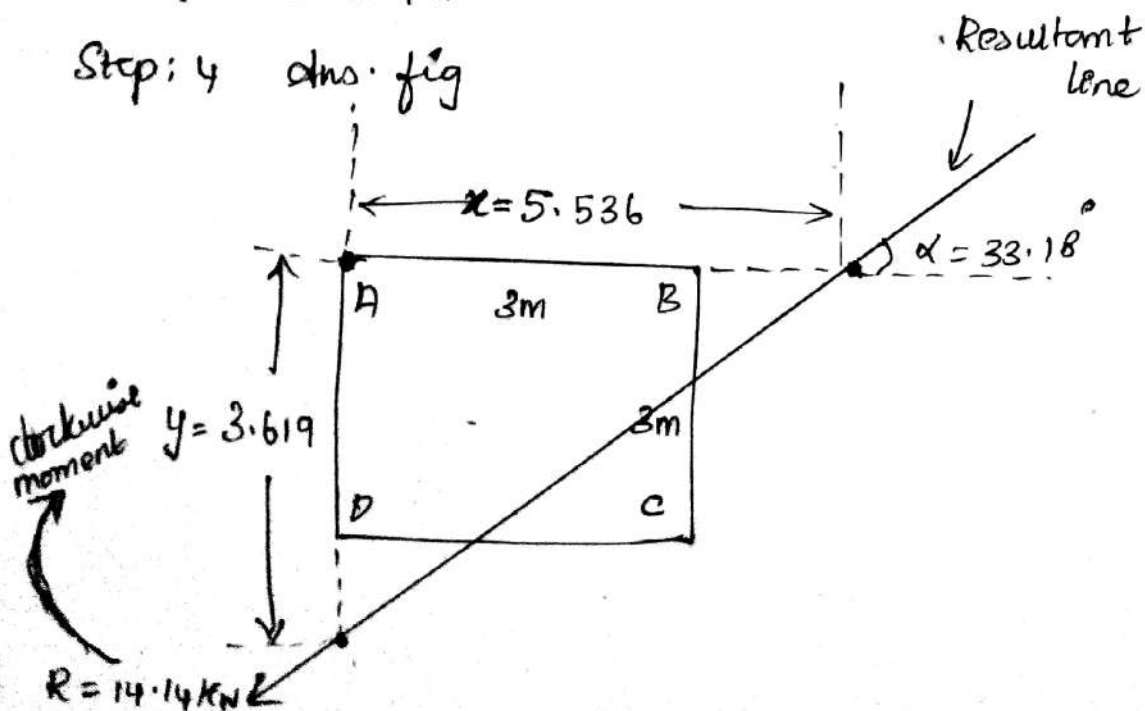
$$x = 5.536 \text{ m}$$

$$y = \left| \frac{\Sigma M_A}{\Sigma F_h} \right|$$

$$= \left| \frac{42.86}{11.84} \right|$$

$$y = 3.619 \text{ m}$$

Step: 4 Ans. fig

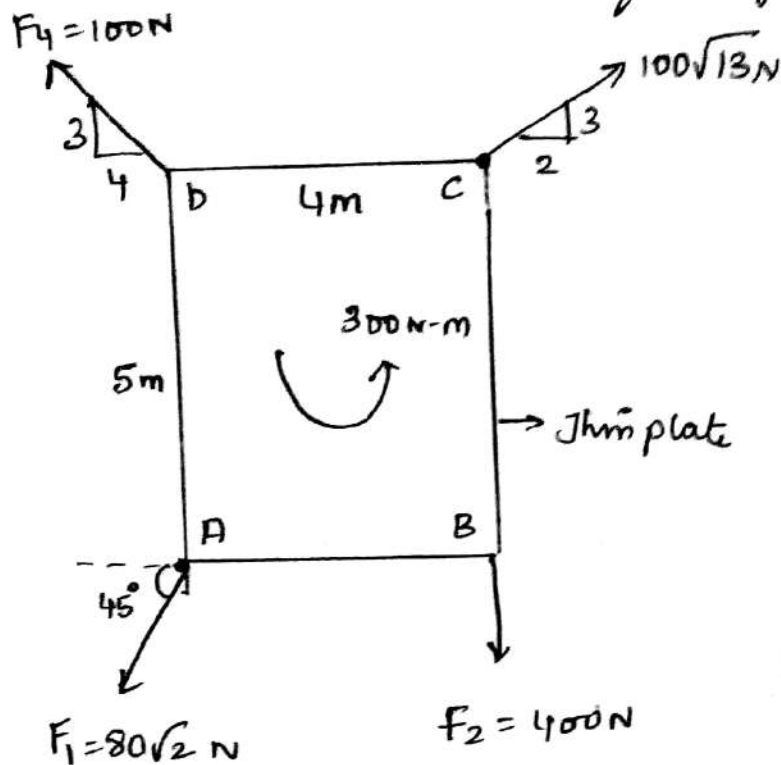


Note:

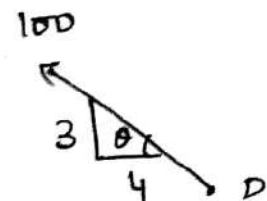
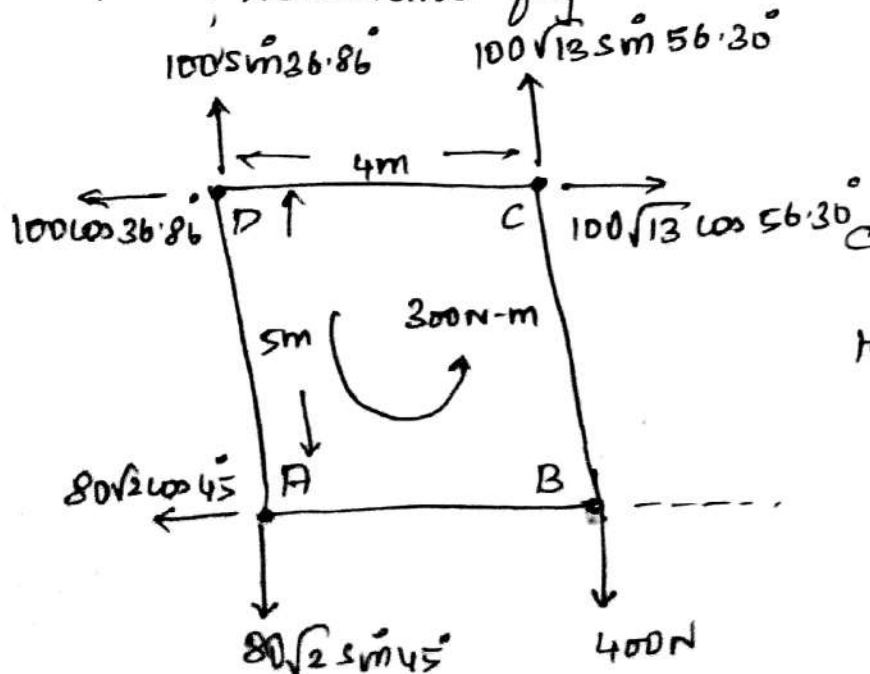
Arrow marks of Resultant force depends on sign of moment about 'A' point.

If $\Sigma M_A = +ve$ (clockwise) R should give clockwise moment about 'A' point.

2. find the magnitude of resultant force and locate the resultant force from A.

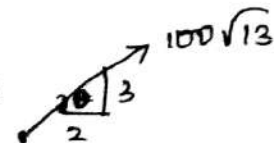


Step: 1, Resultant fig



$$\tan \theta = 3/4$$

$$\theta = 36.86^\circ$$



$$\tan \theta = 3/2$$

$$\theta = 56.30^\circ$$

$$\begin{aligned} \sum F_y &= 100 \sin 36.86^\circ + 100\sqrt{13} \sin 56.30^\circ - 400 \\ &\quad - 80\sqrt{2} \sin 45^\circ \\ &= -120 \text{ N} \end{aligned}$$

$$\sum F_h = -100 \cos 36.86^\circ + 100\sqrt{13} \cos 56.30^\circ - 80\sqrt{2} \cos 45^\circ$$

$$\boxed{\sum F_h = 40 \text{ N}}$$

$$R = \sqrt{\sum F_v^2 + \sum F_h^2}$$

$$= \sqrt{(-120)^2 + 40^2}$$

$$\boxed{R = 126.49 \text{ N}}$$

$$\alpha = \tan^{-1} \left| \frac{\sum F_v}{\sum F_h} \right|$$

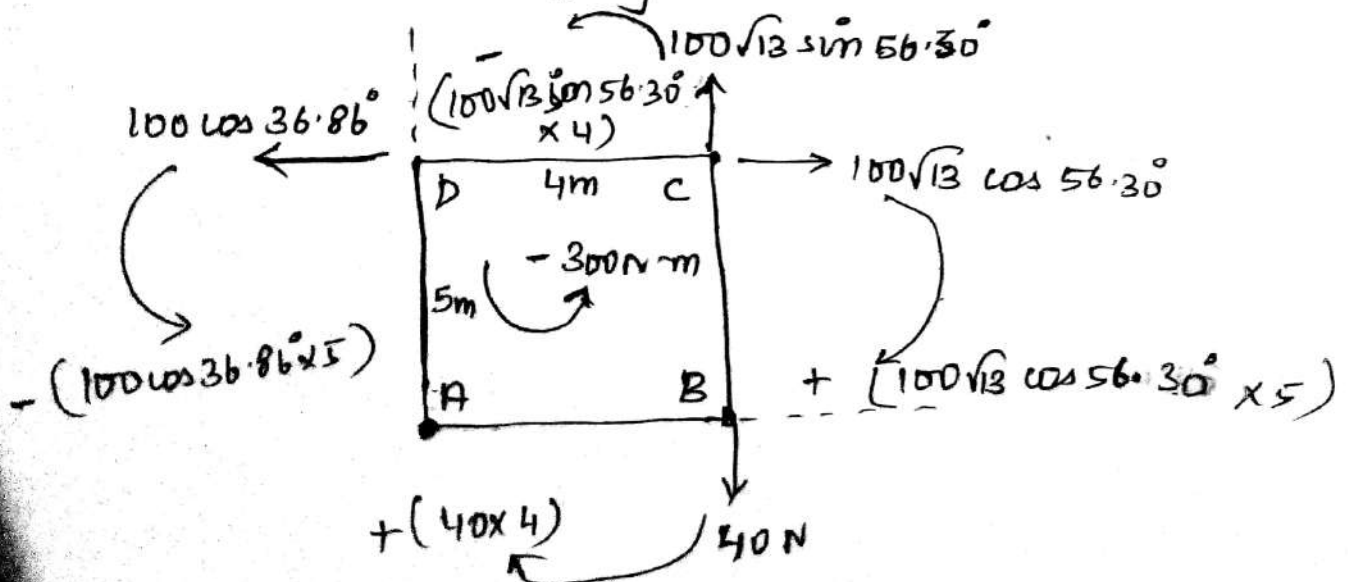
$$= \tan^{-1} \left| \frac{120}{40} \right|$$

$$\boxed{\alpha = 71.56^\circ}$$

Step: 2 finding $\sum M_A = ?$

$$\sum M_A = \sum F_v \times x$$

$$\sum M_A = \sum F_h \times y$$



$$\Sigma M_A = -(100 \cos 36.86^\circ \times 5) - (100 \sqrt{13} \sin 56.30^\circ \times 4) \\ + (100 \sqrt{13} \cos 56.30^\circ \times 5) + (400 \times 4) - 300$$

$$\Sigma M_A = 700.69$$

Step: 3 finding x & y

$$x = \left| \frac{\Sigma M_A}{\Sigma F_v} \right|$$

$$= \left| \frac{60.45}{120} \right|$$

$$x = 0.50 \text{ m}$$

$$x = 5.839 \text{ m}$$

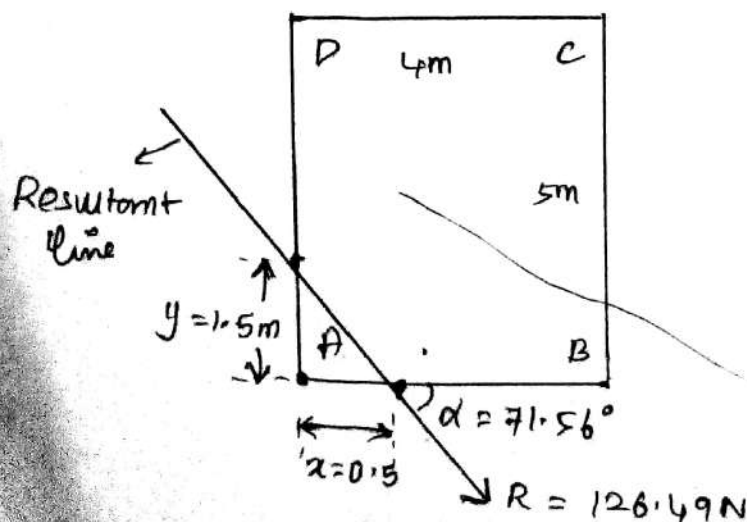
$$y = \left| \frac{\Sigma M_A}{\Sigma F_h} \right|$$

$$= \left| \frac{60.45}{40} \right|$$

$$y = 1.5 \text{ m}$$

$$y = 17.50 \text{ m}$$

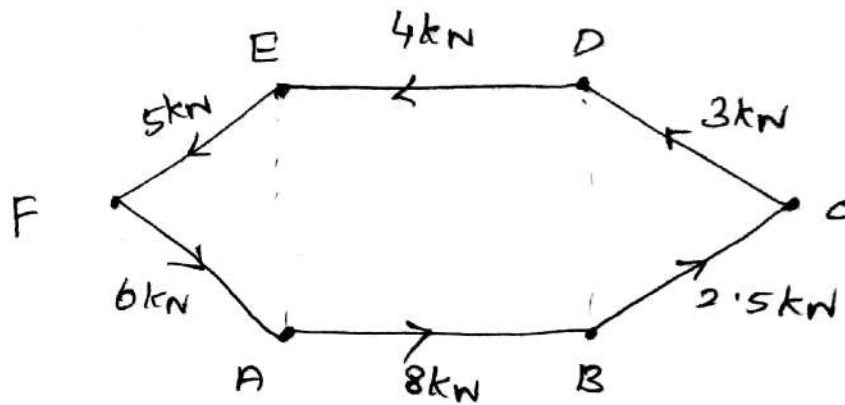
Step: 4 draw fig



Note:

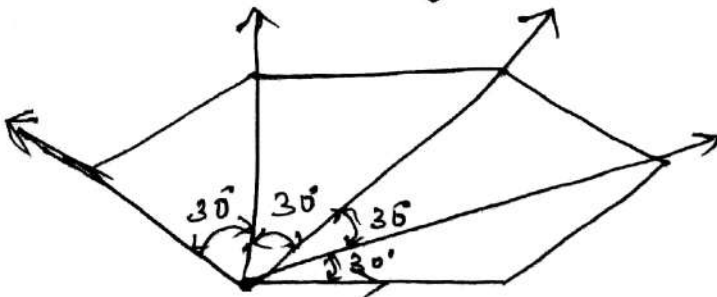
[The given moment is taken as such as in question. No need to multiply with distance. It's already in kn-m]

10. Force acting along the members of the hexagon ABCDEF of side 40 cm as shown in fig. Determine the net moment about A.

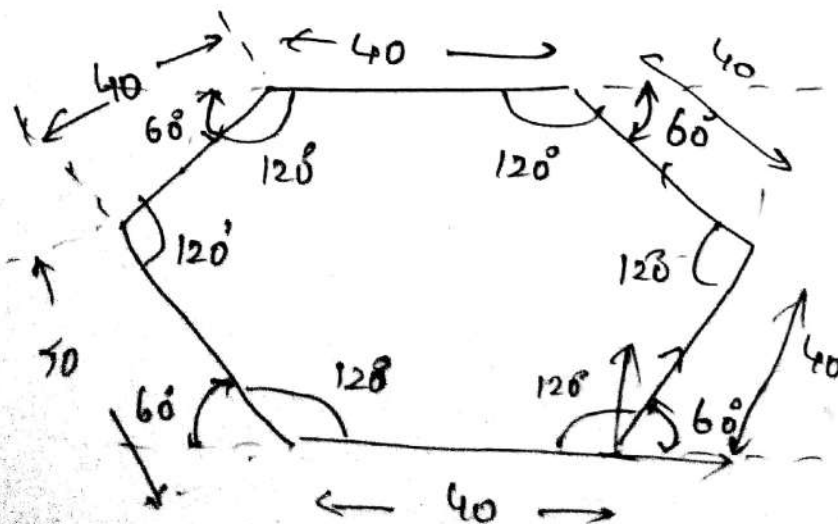


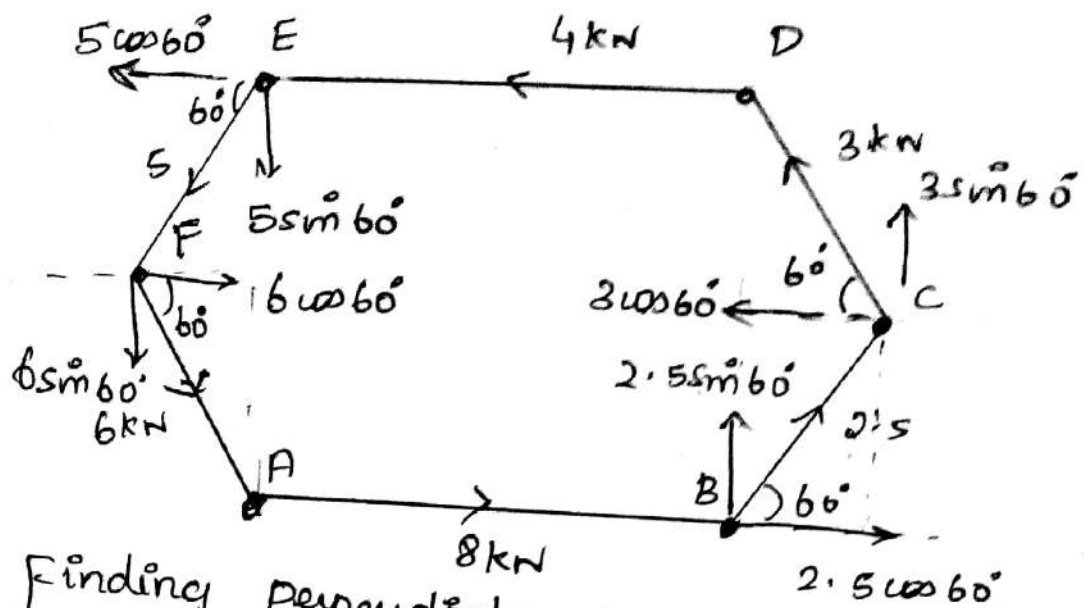
Soln:

Included angle in hexagon $\gamma = \frac{180}{6} = 30^\circ$



Included angle = 120°



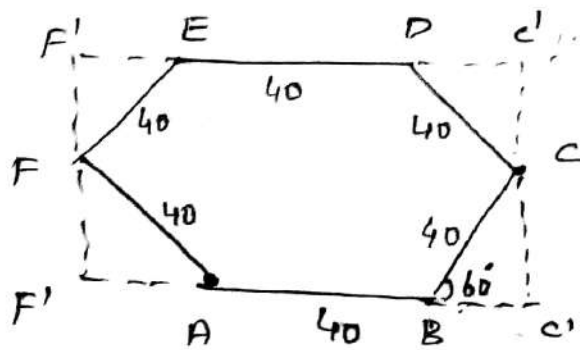
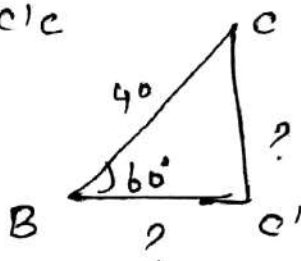


① Finding perpendicular distance

$$BC' = C'D = EF' = F'A$$

$$CC' = FF'$$

$\Delta BC'C$



$$\sin 60^\circ = \frac{CC'}{40} = 34.64, \quad \cos 60^\circ = \frac{BC'}{BC}$$

$$CC' = 34.64 \text{ cm}$$

$$BC' = 20 \text{ cm}$$

② Finding magnitude and direction of resultant force

$$\begin{aligned} \sum F_v &= -5\sin 60^\circ + 3\sin 60^\circ + 2.5\sin 60^\circ - 6\sin 60^\circ \\ &= -4.763 \text{ kN} \end{aligned}$$

$$\begin{aligned} \sum F_h &= -5\cos 60^\circ - 4 - 3\cos 60^\circ + 2.5\cos 60^\circ + 8 \\ &\quad + 6\cos 60^\circ \end{aligned}$$

$$\sum F_h = 4.25 \text{ kN}$$

$$R = \sqrt{\sum F_v^2 + \sum F_h^2}$$

$$= \sqrt{(-4.763)^2 + (4.25)^2}$$

$$R = 6.38 \text{ kN}$$

$$\alpha = \tan^{-1} \left| \frac{\sum F_v}{\sum F_h} \right|$$

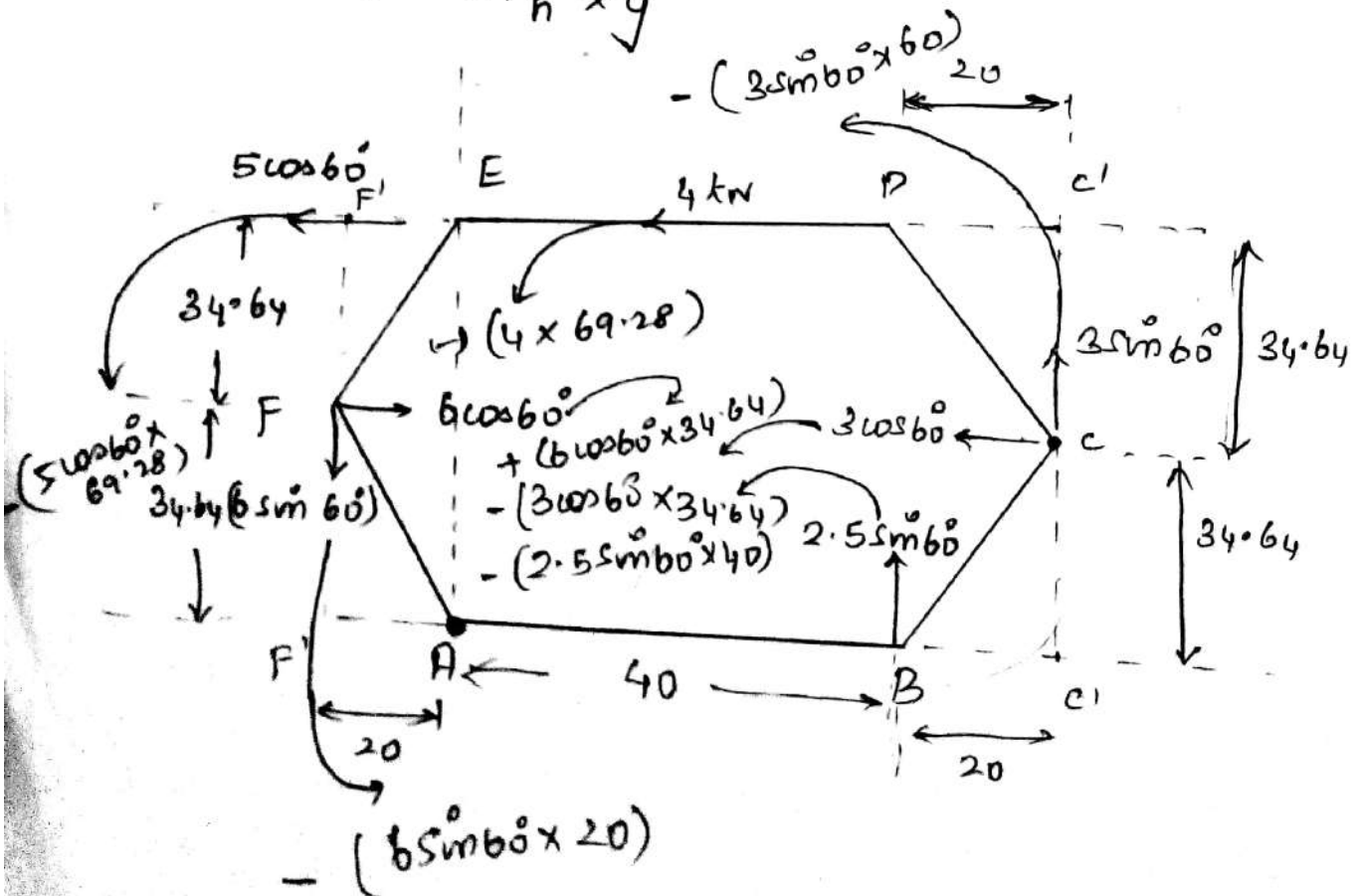
$$= \text{cm}^{-1} \left| \frac{4.763}{4.25} \right|$$

$$\boxed{\alpha = 48.25^\circ}$$

3. Findung SMA: ?

$$S_{MA} = \sum F_V \times x$$

$$\Sigma M_A = \Sigma F_h \times y$$



$$\begin{aligned}\sum M_A = & -(5 \cos 60^\circ \times 69.28) - (3 \sin 60^\circ \times 60) \\ & - (4 \times 69.28) + (6 \cos 60^\circ \times 34.64) \\ & - (3 \cos 60^\circ \times 34.64) - (2.5 \sin 60^\circ \times 40) \\ & - (6 \sin 60^\circ \times 20)\end{aligned}$$

$$\sum M_A = -744.77 \text{ kN-m (Anticlockwise)}$$

4. finding x & y

$$x = \left| \frac{\sum M_A}{\sum F_v} \right|$$

$$y = \left| \frac{\sum M_A}{\sum F_h} \right|$$

$$x = \left| \frac{744.77}{4.763} \right|$$

$$y = \left| \frac{744.77}{4.25} \right|$$

$$x = 156.36 \text{ cm}$$

$$y = 175.24 \text{ cm}$$

5. Ans. $R_F = 6.38 \text{ kN}$

